ADVANCED OLYMPIAD SERIES

Olympiad Combinatorics: Problems and Archive Guide

26 reviewed case-study problems, coverage report, and complete candidate index

Prepared for haohao

Research edition ▪ 5 November 2026

Definitions. Certificates. Complete arguments.

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How to use this file

The first part gives the 26 reviewed case-study problems without hints or answers. Their solutions are in the separate worked-solutions volume. The second part reports the actual archive coverage. The final part indexes every retrieved candidate.

The full text of the included source problems is in the separate `Collected_Problems.pdf` and in `Collected_Source_Problems.md` in the companion ZIP, with one file per problem in `problems/`. Index-only records are not complete problem statements. Imported source solutions are separate from the reviewed solutions.

This is a partial research collection, not the complete historical corpus requested. The coverage report identifies the missing work explicitly.

W01. A gcd–lcm process

Source: IMO 2026, Problem 1; official paper. Mixed number theory and combinatorics.

Begin with 2026 integers greater than one. A move chooses two current entries $a, b > 1$ and replaces them by $g = \gcd(a, b)$ and $\operatorname{lcm}(a, b)/g$. Prove that play must end with exactly one entry greater than one, and that this final entry is independent of all choices.

W02. Two covering counts have the same parity

Source: RMM 2012, Problem 1. **Record:** `rmm_2012_b79110`.

In a finite bipartite acquaintance graph with parts L, R, a subset of either part covers the other part when every vertex of the other part has a neighbor in it. Prove that the numbers of covering subsets on the two sides have the same parity.

W03. Edges above the bipartite threshold force triangles

Source: APMO 1989, Problem 4. **Record:** `apm_1989_ab9780`.

A simple graph has n vertices and m edges. Prove that its number t of triangles satisfies

$$t \geq \frac{4m(m - n^2/4)}{3n}.$$

W04. One illuminated square

Source: APMO 2007, Problem 5. **Record:** `apm_2007_1337e6`.

On a 5×5 board, a switch toggles its own light and its orthogonally adjacent lights. Starting with all lights off, which positions can be the only illuminated square?

W05. An odd grid and a saturated count

Source: India IMO TST 2016, Problem 5. **Record:** `ind_2016_9a166a`.

Color each edge of an $n \times n$ grid red or blue, where n is odd. If at most n^2 edges are red, prove that a unit square has at least three blue edges.

W06. An odd route with distinct airlines

Source: RMM 2020, Problem 3. **Record:** `rou_2020_2530c6`.

An n-vertex flight network has n airline colors. Each color supports an odd closed route. Prove that an odd closed route exists whose edges have pairwise different colors.

W07. A convex-hull identity

Source: IMO Shortlist 2006, C3. **Record:** `imo_2006_f50971`.

Let S be a finite planar set with no three collinear points. For a convex polygon P with vertices in S, let $a(P)$ count its vertices and $b(P)$ the points of S outside it. Include segments, points and the empty polygon. Prove for every real x that

$$\sum_{P} x^{a(P)}(1-x)^{b(P)} = 1.$$

W08. Tournaments generated by commuting involutions

Source: IMO Shortlist 2006, C5. **Record:** `imo_2006_b88369`.

In each of k rounds, n players are paired, with no pair repeated. Require that whenever A, B and C, D are pairs in one round and A, C is a pair in another, then B, D is also a pair in that other round. Determine the possible positive integers (n, k).

W09. Synchronized bishops

Source: RMM 2024, Problem 1. **Record:** `rou_2024_d65b67`.

Put one labeled bishop in each square of the top row of a $2^n \times 2^n$ board. A jump moves every bishop diagonally so that they again occupy distinct squares in one common row. How many orders can occur on the bottom row?

W10. Preventing a directed cycle forever

Source: RMM 2018, Problem 3. **Record:** `rou_2018_a59bad`.

Ann and Bob alternately orient previously unoriented unit edges of the infinite square grid, with Ann first. Bob wins if a directed cycle ever appears, including after Ann's move. Can he force a win?

W11. A sum-free sequence at the density boundary

Source: USA TSTST 2013, Problem 1, as identified in the recovered selection booklet. **Record:** `usa_2013_bfa49c`.

Find all strictly increasing positive integer sequences (a_i) with no equation $a_i + a_j = a_k$, allowing repeated indices, and with $a_k = 2k - 1$ for infinitely many k.

W12. Large disjoint cross-intersecting families

Source: IMO Shortlist 2007, C7. **Record:** `imo_2007_67267d`.

For every $0 < \alpha < (3 - \sqrt{5})/2$, find n, p with $p > \alpha 2^n$ and two disjoint families $\mathcal{S}, \mathcal{T}$, each of p distinct subsets of $[n]$, such that each member of $\mathcal{S}$ intersects every member of $\mathcal{T}$.

W13. Fifteen subsets of a 56-element set

Source: China Mathematical Olympiad 2006, Problem 6, attributed to Leng Gangsong. **Record:** `chn_2006_c22bc5`.

Find the smallest integer N such that, if every seven of fifteen subsets of a 56-element set have union of size at least N, some three subsets have a common element.

W14. Forcing a rainbow three-cell window

Source: RMM 2012, Problem 5. **Record:** `rmm_2012_704e6f`.

An $n \times n$ board, $n \geq 3$, uses exactly $C = \lfloor (n+2)^2/3 \rfloor$ colors. Prove that some horizontal or vertical three-cell window has three different colors.

W15. The anti-Pascal pyramid

Source: IMO 2018, Problem 3; also IMO Shortlist 2018, C4. **Record:** `imo_2018_55614a`.

In a triangular array, every entry above the bottom row equals the absolute difference of its two children. Can an array with 2018 rows contain every integer from 1 through $1 + 2 + \cdots + 2018$ exactly once?

W16. Knights against blocking queens

Source: IMO Shortlist 2018, C2. **Record:** `imo_2018_eed763`.

Horst places a knight on an empty square of a 20×20 board without attacking any previous knight. Queenie then occupies an empty square with a queen. They repeat until a player cannot move. Queens only occupy squares; they impose no attack restriction. What number of knights can Horst guarantee?

W17. One spanning tree meeting three quotas

Source: RMM 2023, Problem 6. **Record:** `rou_2023_0cea08`.

A connected graph has $r + g + b + 1$ vertices, with red, green and blue edges. There is a spanning tree with exactly r red edges, another with exactly g green edges, and another with exactly b blue edges. Prove that one tree meets all three quotas simultaneously.

W18. Keeping 150 entries of a doubly stochastic matrix

Source: China TST 2023, Test Three, Day 2, Problem 6. **Record:** `chn_2023_a44cf5`.

In any nonnegative 100×100 matrix with every row sum and column sum equal to one, retain 150 entries and set the others to zero. What is the largest universal lower bound λ on all remaining row and column sums?

W19. A set whose largest half is constrained

Source: USAMO 2006, Problem 2. **Record:** `usa_2006_9454b3`.

For a positive integer k, find the least N for which $2k+1$ distinct positive integers have total sum greater than N, while every k-element subset has sum at most $N/2$.

W20. Unique subsums with odd parts

Source: India IMO TST 2013, Problem 3. **Record:** `ind_2013_5ef73e`.

A partition of 9999 into odd positive parts is called sum-friendly if each integer from 0 to 9999 has exactly one subsum representation. Equal copies of a part are indistinguishable: a representation specifies only how many copies of each distinct value are used. Count the sum-friendly partitions.

W21. Equal totals but strongly unbalanced rows and columns

Source: Vietnam Mathematical Olympiad 2016, Day 1, Problem 4. **Record:** `vnm_2016_c11c43`.

An $m \times n$ board contains equal numbers of two symbols. In each row their counts differ by at least $n/2$, and in each column by at least $m/2$. Show that m, n are divisible by four, and construct such a board for $m = n = 2016$.

W22. Cookie pebbling on an even cycle

Source: Korean Mathematical Olympiad 2010, Problem 5. National pipeline supplement. **Record:** `kor_2010_1dc4be`.

At the vertices of a cycle with $2n$ vertices are nonnegative integer numbers of cookies. A move consumes two cookies at a vertex and places one at a neighboring vertex. Find the smallest total number guaranteeing that a prescribed vertex can obtain a cookie, regardless of the initial distribution.

W23. A finite state space with draws

Source: Balkan Mathematical Olympiad 2018, Problem 3. **Record:** `mkd_2018_3ac94a`.

Starting from positive piles (a, b), Alice and Bob alternate, Alice first. A move chooses an even pile and transfers half of it to the other pile. A player with no legal move loses. Determine exactly when Bob can force a win.

W24. The fewest tiles in a maximal placement

Source: EGMO 2016, Problem 5. **Record:** `rou_2016_9aa109`.

Let $k \geq 2$ and $k \leq n \leq 2k-1$. Place nonoverlapping $1 \times k$ or $k \times 1$ tiles on an $n \times n$ board until no further tile fits. What is the smallest possible number of tiles?

W25. A private witness in every proper-union subfamily

Source: RMM 2017, Problem 3. **Record:** `rou_2017_f8413b`.

Let $|X| = n > 1$. A family of distinct proper nonempty subsets of X has the following property: every nonempty subfamily whose union is not X has an element belonging to exactly one member of that subfamily. Find its maximum size.

W26. Reducing a set by sum-and-difference moves

Source: RMM 2021, Problem 4. **Record:** `rmm_2021_03b2ae`.

Start with $\{1, 2, \dots, n\}$, $n \geq 2$. Erase two distinct current numbers a, b, and insert $a + b$ and $|a - b|$, suppressing any duplicate with an existing number. Can the set be reduced to exactly two elements?

Coverage, provenance, and unfinished work

Research edition: 5 November 2026. **This is not a complete century-wide collection, and not every imported solution has been independently solved or checked.**

Exact deliverable counts

- 1,116 retrieved MathNet candidate records, plus one additional official 2026 problem in the reviewed set.
- 1,104 distinct statement groups after whitespace-only exact normalization. Translation and semantic duplicates remain; this is not a unique-problem count.
- 770 source statement files included after scope, rights and figure filters.
- 632 files containing 786 source-supplied solution texts. These are **not independently reviewed** except where an explicit W-number points to the separate rewritten proof.
- 26 reviewed, rewritten case studies, each in its own problem and solution file; 25 correspond to retrieved records and W01 is additional.
- 20 transfer exercises with solutions in the theory book.

Candidate scope categories: contest: 464, shortlist: 233, selection: 205, quarantine: 84, pipeline_supplement: 130.

Definition of the target

The requested century is treated as calendar years 1927–2026, through the research date. A competition's years before it existed are not missing papers. However, an authoritative founding/cancellation calendar was not reconstructed for every selected competition, so this edition does not label every blank year "not held." No authoritative ranking of the ten hardest competitions exists. The editorial contest set is IMO, RMM, CMO-China, USAMO, ARMO, IranMO, VMO, APMO, Balkan MO and EGMO. Difficulty was not measured problem by problem.

The country set follows the official 2026 team order: China, United States, Russia, Singapore, Vietnam, South Korea, India, North Korea, United Kingdom and Japan. This is a transparent recent-year convention, not an all-time top-ten ranking. Source: https://www.imo-official.org/results/team/year/2026/ .

National contests, camps, shortlists, team-selection exams and the actual international paper are different objects. The ledger preserves the source label, a corrected label where a contradiction could be resolved, and a scope status. It never treats the number of downloaded rows as certified historical coverage.

Recovered years by contest family

Counts below are candidate records, not unique verified problems. A listed year indicates at least one recovered candidate, not a complete paper or complete combinatorics set. W01 (IMO 2026) is additional to this table.

Family	Candidate records	Years with at least one recovered record
IMO	29	1983, 1984, 1985, 1986, 1987, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2012, 2013, 2014, 2016
RMM	38	2011, 2012, 2013, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2023, 2024, 2025
CMO-China	39	2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2022, 2023, 2024
USAMO	33	2003, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2021, 2025
ARMO	41	2010, 2019, 2025
IranMO	141	2010, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025
VMO	48	2001, 2002, 2003, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2018, 2019, 2020, 2021, 2022, 2023, 2024
APMO	36	1989, 1990, 1991, 1992, 1993, 1998, 1999, 2002, 2003, 2004, 2005, 2007, 2008, 2009, 2010, 2016, 2018, 2019, 2020, 2022, 2023, 2024, 2025
BMO	32	2004, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2018, 2019, 2020, 2023
EGMO	27	2012, 2013, 2014, 2015, 2016, 2018, 2020, 2023, 2024, 2025
IMO-SL	168	2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024
BMO-SL	65	2010, 2011, 2012, 2014, 2015, 2016, 2017, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Recovered national selection coverage

Only records explicitly classified as selection are counted in the second column. Pipeline supplements are separately counted and do not fill a final-selection gap.

Country	Selection candidates	Pipeline supplements	Recovered selection years
China	77	0	2007, 2008, 2009, 2010, 2011, 2013, 2014, 2021, 2022, 2023, 2024, 2025
United States	77	0	2001, 2003, 2004, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2018, 2019, 2020, 2023, 2025
Russia	0	0	No recovered records
Singapore	4	0	2010
Vietnam	28	0	2001, 2002, 2003, 2010, 2011, 2012, 2015, 2017, 2018, 2021, 2022, 2023, 2024
South Korea	0	25	No recovered records
India	19	3	2010, 2011, 2012, 2013, 2016, 2023, 2024
North Korea	0	0	No recovered records
United Kingdom	0	0	No recovered records
Japan	0	102	No recovered records

What is still missing

1. An authoritative per-year, per-round inventory for all ten competitions and all ten countries, including founding years, cancellations and unpublished selection tests.
2. Most early historical material: the downloaded candidates range from 1983 to 2025; the reviewed set adds 2026. This does not cover 1927–1982, nor every year after 1983.
3. Complete final-selection archives for Russia, South Korea, North Korea, the United Kingdom and Japan; only some national pipeline material or source pointers were recovered for several of them. Singapore coverage is very small.
4. Original-paper verification of every contest/year/problem attribution. Several source normalized codes were plainly wrong and have been corrected or quarantined.
5. Semantic deduplication across translations and reused selection problems. The UK archive expressly notes reuse, making this essential.
6. Independent solutions to every included problem, and independent review of every supplied solution. The verified-exposition set is W01–W26 only.
7. Restoration of external figures and any necessary permissions for source-text reproduction. Index-only records are not complete problem statements.
8. A theory synthesis based on the entire eventual corpus. This book derives its case-based synthesis from 26 reviewed problems and classical results, not all 1,116 downloaded records.

Provenance and reproduction policy

The principal retrieval source is MathNet's website snapshot `312a167f6df5`, built 29 April 2026. Website and Hugging Face snapshots have different record counts and metadata schemas; they were not

assumed interchangeable. The recovered website records preserve source booklets, authors, organization, language, rights fields, original competition strings and problem numbers where provided.

MathNet releases records under CC BY 4.0 where no explicit original rights are asserted, and says original rights assertions take precedence. This package conservatively withholds raw statements and raw solutions whenever the recovered record contains an explicit holder or license restriction. Such records retain bibliographic metadata and a source link. A missing rights field is not an independent legal investigation; the inclusion policy relies on MathNet's stated fallback license. Our restatements and rewritten mathematical expositions do not reproduce the withheld source prose.

Records requiring external images are not silently presented as complete. Figure-dependent source solutions are omitted while other self-contained solutions may be included. English translation was not performed across the whole corpus; source languages are retained.

All source-supplied proofs retain their status. Source OCR and model-assisted extraction can contain incorrect signs, formulas, numbering, translations and attributions. The audit's known errata are examples, not a warranty that no other errors remain. Another public IMO dataset was excluded after an algebraically false supplied argument was found; its proofs were not imported.

Source register

- MathNet project and paper: https://mathnet.mit.edu/
- MathNet dataset and license statement: https://huggingface.co/datasets/ShadenA/MathNet
- CC BY 4.0: https://creativecommons.org/licenses/by/4.0/
- Official IMO papers: https://www.imo-official.org/problems/
- Official 2026 IMO paper: https://www.imo-official.org/assets/documents/problems/2026/2026_eng.pdf
- RMM organizer archive, example year: https://rmms.lbi.ro/rmm2024/index.php?id=problems_math
- RMM historical links: https://bmos.ukmt.org.uk/home/rmm.shtml
- APMO official archive: https://www.apmo-official.org/problems
- EGMO official archive: https://www.egmo.org/
- USA TST archive: https://github.com/usa-tst-public/USA-TST-Archive
- UK selection tests: https://bmos.ukmt.org.uk/home/selection-tests.shtml

MathNet citation: Shaden Alshammari, Kevin Wen, Abrar Zainal, Mark Hamilton, Navid Safaei, Sultan Albarakati, William T. Freeman and Antonio Torralba, *MathNet: A Global Multimodal Benchmark for Mathematical Reasoning and Retrieval*, ICLR 2026. Original contest organizations and source authors remain credited in each record where the dataset supplies them.

Completion status against the four requested tasks

Task	Status in this edition
Gather all historical combinatorics problems	Partial research archive; substantial missing years, rounds and figure/rights gaps
Solve all problems	26 checked rewritten solutions; additional supplied solutions retained without independent verification
Read all solutions and derive theories	Synthesis from the reviewed set and classical theory; not an all-record review

Task	Status in this edition
Write a rigorous preparation book	Completed research-edition manuscript with proved tools, worked-case links and solved exercises; not a claim of encyclopedic completeness or external peer review

Candidate index

Each entry is a source record, not necessarily a unique or correctly attributed historical problem. The CSV and JSON ledger contain full metadata, source links and classifications. “Text” means a statement file is included; “pointer” means use the ledger source link. “S” counts supplied source solutions included in this package, not checked proofs.

APMO

- `apm_1989_ab9780` — 1989; 4; contest; text; S=1; checked W03.
- `apm_1990_c222d1` — 1990; 4; contest; text; S=1.
- `apm_1990_e66529` — 1990; 2; contest; text; S=1.
- `apm_1991_b65647` — 1991; 2; contest; text; S=1.
- `apm_1992_0e43a2` — 1992; 4; contest; text; S=1.
- `apm_1993_2c32ee` — 1993; 5; contest; text; S=1.
- `apm_1998_69a314` — 1998; 1; contest; text; S=1.
- `apm_1999_563b73` — 1999; 5; contest; text; S=0.
- `apm_1999_d63d04` — 1999; 2; contest; text; S=1.
- `apm_2002_ee5012` — 2002; 1; contest; text; S=2.
- `apm_2003_d3281e` — 2003; 5; contest; text; S=1.
- `apm_2004_1843da` — 2004; 1; contest; text; S=1.
- `apm_2004_f5148e` — 2004; 3; contest; text; S=0.
- `apm_2005_be8587` — 2005; Problem 4; contest; text; S=0.
- `apm_2007_1337e6` — 2007; Problem 5; contest; text; S=1; checked W04.
- `apm_2007_7a497c` — 2007; Problem 3; contest; text; S=1.
- `apm_2007_ca2bc8` — 2007; Problem 1; contest; text; S=2.
- `apm_2008_44c90f` — 2008; Problem 2; contest; text; S=1.
- `apm_2008_ec38c0` — 2008; Problem 4; contest; text; S=1.
- `apm_2009_dd163e` — 2009; Problem 1; contest; text; S=1.
- `apm_2010_bde43c` — 2010; 3; contest; text; S=3.
- `apm_2016_6dfebe` — 2016; 4; contest; text; S=1.
- `mkd_2018_77a86a` — 2018; 3; contest; text; S=0.
- `apm_2019_3a3563` — 2019; 4; contest; text; S=0.
- `apm_2020_2b1cf6` — 2020; 4; contest; text; S=1.
- `apm_2020_9793f2` — 2020; 2; contest; text; S=2.
- `apm_2020_f3d3c1` — 2020; 3; contest; text; S=2.
- `apm_2022_04f0c2` — 2022; 4; contest; text; S=1.
- `apm_2023_44bc62` — 2023; 4; contest; text; S=1.
- `apm_2023_effff8` — 2023; 5; contest; text; S=0.
- `apm_2023_fb45d1` — 2023; 1; contest; text; S=0.
- `apm_2024_299020` — 2024; 4; contest; text; S=1.
- `apm_2024_531576` — 2024; 3; contest; text; S=1.
- `apm_2024_91e126` — 2024; 2; contest; text; S=1.
- `apm_2025_0453f0` — 2025; 2; contest; text; S=1.
- `apm_2025_e1a993` — 2025; 4; contest; text; S=3.

ARMO

- rus_2010_03f505 — 2010; 11.2.; contest; pointer; S=0.
- rus_2010_3f9675 — 2010; 10.4.; contest; pointer; S=0.
- rus_2010_52761d — 2010; 9.7.; contest; pointer; S=0.
- rus_2010_52dfd4 — 2010; 9.2.; contest; pointer; S=0.
- rus_2010_6d1604 — 2010; 10.8.; contest; pointer; S=0.
- rus_2010_731624 — 2010; 10.2.; contest; pointer; S=0.
- rus_2010_73d9a8 — 2010; 9.5.; contest; pointer; S=0.
- rus_2010_a5a493 — 2010; 10.1.; contest; pointer; S=0.
- rus_2010_b41884 — 2010; 9.6.; contest; pointer; S=0.
- rus_2010_b9e241 — 2010; 10.1.; contest; pointer; S=0.
- rus_2010_c237cf — 2010; 11.2.; contest; pointer; S=0.
- rus_2010_c43958 — 2010; 10.8.; contest; pointer; S=0.
- rus_2010_c8b08d — 2010; 11.8.; contest; pointer; S=0.
- rus_2010_d3c87f — 2010; 11.5.; contest; pointer; S=0.
- rus_2010_ff78d5 — 2010; 9.4.; contest; pointer; S=0.
- rus_2019_00e8c3 — 2019; 11.10; contest; pointer; S=0.
- rus_2019_2b25cd — 2019; 9.7; contest; pointer; S=0.
- rus_2019_2b53d8 — 2019; 9.4; contest; pointer; S=0.
- rus_2019_3dae1d — 2019; 11.8; contest; pointer; S=0.
- rus_2019_41fd0c — 2019; 9.2; contest; pointer; S=0.
- rus_2019_519d9d — 2019; 11.3; contest; pointer; S=0.
- rus_2019_5d1760 — 2019; 10.3; contest; pointer; S=0.
- rus_2019_618af1 — 2019; 11.3; contest; pointer; S=0.
- rus_2019_818166 — 2019; 9.10; contest; pointer; S=0.
- rus_2019_927fb5 — 2019; 9.5; contest; pointer; S=0.
- rus_2019_928cad — 2019; 9.7; contest; pointer; S=0.
- rus_2019_9c74b3 — 2019; 10.3; contest; pointer; S=0.
- rus_2019_b5cc6a — 2019; 10.2; contest; pointer; S=0.
- rus_2019_d54f19 — 2019; 9.9; contest; pointer; S=0.
- rus_2019_d99efd — 2019; 10.7; contest; pointer; S=0.
- rus_2025_2d96d8 — 2025; 11.3; contest; pointer; S=0.
- rus_2025_49ced6 — 2025; 9.7; contest; pointer; S=0.
- rus_2025_522ea4 — 2025; 11.4; contest; pointer; S=0.
- rus_2025_6aa49c — 2025; 10.7; contest; pointer; S=0.
- rus_2025_7a4604 — 2025; 10.1; contest; pointer; S=0.
- rus_2025_8671a6 — 2025; 11.6; contest; pointer; S=0.
- rus_2025_92724f — 2025; 10.4; contest; pointer; S=0.
- rus_2025_95c4c3 — 2025; 11.1; contest; pointer; S=0.
- rus_2025_bafd00 — 2025; 9.6; contest; pointer; S=0.
- rus_2025_cd1fbb — 2025; 9.4; contest; pointer; S=0.
- rus_2025_d73fb6 — 2025; 9.1; contest; pointer; S=0.

BMO

- rou_2004_775bce — 2004; 4.; contest; text; S=0.
- rou_2004_a43add — 2004; Problema 3; contest; text; S=0.
- grc_2007_31776c — 2007; 4; contest; text; S=2.

- mkd_2008_30218d — 2008; 3; contest; text; S=1.
- mkd_2008_8fbac0 — 2008; 3; contest; text; S=1.
- mkd_2008_f3b3c2 — 2008; 4; contest; text; S=1.
- bmo_2009_f956f2 — 2009; 3; contest; text; S=0.
- mkd_2009_24150b — 2009; 2??.; contest; text; S=1.
- mkd_2009_5131ff — 2009; 4?.; contest; text; S=1.
- mkd_2009_620603 — 2009; 3?.; contest; text; S=1.
- mkd_2009_895f98 — 2009; 3.; contest; text; S=1.
- mkd_2009_c6e2af — 2009; 3.; contest; text; S=1.
- mkd_2010_b41ce7 — 2010; Problem 3.; contest; text; S=1.
- rou_2010_b512fc — 2010; Problem 3.; contest; text; S=0.
- mkd_2011_c1004b — 2011; 3; contest; text; S=1.
- mkd_2011_c92f45 — 2011; 5; contest; pointer; S=1.
- mkd_2012_683110 — 2012; 3; contest; text; S=2.
- bmo_2013_654e0d — 2013; 7; contest; text; S=1.
- bmo_2013_84acf9 — 2013; 2; contest; text; S=1.
- bmo_2013_a11bcd — 2013; 4; contest; text; S=1.
- bmo_2013_a923a3 — 2013; 5; contest; text; S=1.
- grc_2013_323f49 — 2013; Problem 4.; contest; text; S=0.
- rou_2013_306b97 — 2013; Problem 4; contest; text; S=0.
- mkd_2014_9ea0c9 — 2014; 4; contest; text; S=0.
- bmo_2015_0b84eb — 2015; NT5.; contest; text; S=1.
- bmo_2015_2acf30 — 2015; NT2.; contest; text; S=1.
- bmo_2015_6abd41 — 2015; NT1.; contest; text; S=1.
- mkd_2018_3ac94a — 2018; 3; contest; text; S=1; checked W23.
- grc_2019_037737 — 2019; Problem 4.; contest; text; S=1.
- rou_2020_3e985a — 2020; Problem 3.; contest; pointer; S=0.
- rou_2020_ed3b79 — 2020; Problem 2.; contest; pointer; S=0.
- jbm_2023_78057b — 2023; Problem 3.; contest; text; S=1.

BMO-SL

- bmo_2010_2df1d0 — 2010; NT1; shortlist; text; S=1.
- bmo_2010_36a58c — 2010; C1; shortlist; text; S=1.
- bmo_2010_76e4e6 — 2010; C4; shortlist; text; S=1.
- bmo_2010_82e1d0 — 2010; C2; shortlist; text; S=1.
- bmo_2010_897051 — 2010; A4; shortlist; text; S=1.
- bmo_2010_8eb8ae — 2010; C5; shortlist; text; S=1.
- bmo_2011_2d432f — 2011; C3; shortlist; text; S=1.
- bmo_2012_e19c1d — 2012; C1; shortlist; text; S=2.
- bmo_2014_5daa9c — 2014; C2; shortlist; text; S=1.
- bmo_2014_64e0c9 — 2014; A5; shortlist; text; S=1.
- bmo_2014_96bc5b — 2014; C3; shortlist; text; S=0.
- bmo_2014_bde1ed — 2014; C1; shortlist; text; S=1.
- bmo_2015_5a0b4c — 2015; C2.; shortlist; text; S=1.
- bmo_2015_9ee8d6 — 2015; C1.; shortlist; text; S=1.
- bmo_2015_e3a469 — 2015; C3.; shortlist; text; S=1.
- bmo_2016_6d4fd5 — 2016; C1.; shortlist; text; S=1.
- bmo_2016_715bc2 — 2016; N5.; shortlist; text; S=1.

- bmo_2016_82a6cb — 2016; C3.; shortlist; text; S=0.
- bmo_2016_bcac10 — 2016; C2.; shortlist; text; S=1.
- bmo_2017_386ab9 — 2017; C1; shortlist; text; S=1.
- bmo_2017_3e2f81 — 2017; C2; shortlist; text; S=1.
- bmo_2017_3f4864 — 2017; C3; shortlist; text; S=1.
- bmo_2017_5eecca — 2017; C5; shortlist; text; S=1.
- bmo_2017_79742c — 2017; C6; shortlist; text; S=1.
- bmo_2017_7d83b0 — 2017; A3; shortlist; text; S=1.
- bmo_2019_17ff25 — 2019; C1; shortlist; text; S=1.
- bmo_2019_404973 — 2019; TN1; shortlist; text; S=1.
- bmo_2019_52e757 — 2019; C2; shortlist; text; S=1.
- bmo_2019_840a04 — 2019; C3; shortlist; text; S=1.
- bmo_2019_dd840b — 2019; C4; shortlist; text; S=0.
- bmo_2020_2d224f — 2020; C4; shortlist; text; S=1.
- bmo_2020_327627 — 2020; N2; shortlist; text; S=1.
- bmo_2020_dff0bd — 2020; C1; shortlist; text; S=1.
- bmo_2021_085336 — 2021; C5; shortlist; text; S=2.
- bmo_2021_0a2e16 — 2021; C4; shortlist; text; S=1.
- bmo_2021_2fdefd — 2021; A4; shortlist; text; S=2.
- bmo_2021_958307 — 2021; C2; shortlist; text; S=1.
- bmo_2021_ac9fb9 — 2021; N7; shortlist; text; S=1.
- bmo_2021_c13570 — 2021; C6; shortlist; text; S=1.
- bmo_2021_c837cf — 2021; C3; shortlist; text; S=2.
- bmo_2022_2d08c3 — 2022; C2.; shortlist; text; S=1.
- bmo_2022_34f8be — 2022; A5.; shortlist; text; S=1.
- bmo_2022_36e10e — 2022; C5.; shortlist; text; S=2.
- bmo_2022_380ffa — 2022; N2.; shortlist; text; S=2.
- bmo_2022_4d73ec — 2022; A2.; shortlist; text; S=2.
- bmo_2022_8f10e3 — 2022; C4.; shortlist; text; S=0.
- bmo_2022_9167c0 — 2022; C1.; shortlist; text; S=1.
- bmo_2022_977a4b — 2022; N4.; shortlist; text; S=2.
- bmo_2022_9fea53 — 2022; C3.; shortlist; text; S=1.
- bmo_2023_3cf7b2 — 2023; C3; shortlist; text; S=0.
- bmo_2023_4f2ba4 — 2023; C4; shortlist; text; S=1.
- bmo_2023_5e97b2 — 2023; C2; shortlist; text; S=0.
- bmo_2023_aef8b8 — 2023; C1; shortlist; text; S=1.
- bmo_2023_be5fcc — 2023; C5; shortlist; text; S=1.
- bmo_2024_17dc50 — 2024; C1.; shortlist; text; S=1.
- bmo_2024_22c656 — 2024; C5.; shortlist; text; S=1.
- bmo_2024_37ccab — 2024; C3.; shortlist; text; S=1.
- bmo_2024_3c60be — 2024; C4.; shortlist; text; S=2.
- bmo_2024_4b7efa — 2024; C2.; shortlist; text; S=1.
- bmo_2025_012c01 — 2025; C6; shortlist; text; S=0.
- bmo_2025_75a6a6 — 2025; N3; shortlist; text; S=1.
- bmo_2025_8bfc7d — 2025; C2; shortlist; text; S=0.
- bmo_2025_a483fd — 2025; C4; shortlist; text; S=1.
- bmo_2025_b27827 — 2025; C1; shortlist; text; S=1.
- bmo_2025_d22b64 — 2025; C5; shortlist; text; S=3.

CMO-China

- chn_2003_0bb333 — 2003; 2; contest; pointer; S=0.
- chn_2003_b959d3 — 2003; 5; contest; pointer; S=0.
- chn_2004_0d1f01 — 2004; 6; contest; pointer; S=0.
- chn_2004_891c51 — 2004; 3; contest; pointer; S=0.
- chn_2005_34f153 — 2005; 5; contest; pointer; S=0.
- chn_2006_38a222 — 2006; 3; contest; pointer; S=0.
- chn_2006_c22bc5 — 2006; 6; contest; pointer; S=0; checked W13.
- chn_2006_fc775a — 2006; 2; contest; pointer; S=0.
- chn_2007_4036cd — 2007; 3.; contest; pointer; S=0.
- chn_2007_ce1f06 — 2007; 2.; contest; pointer; S=0.
- chn_2007_e90d96 — 2007; 6; contest; pointer; S=0.
- chn_2008_17f19a — 2008; 2; contest; pointer; S=0.
- chn_2008_3cc293 — 2008; 4.; contest; pointer; S=0.
- chn_2008_af53c4 — 2008; 5.; contest; pointer; S=0.
- chn_2008_d26bb7 — 2008; 3.; contest; pointer; S=0.
- chn_2009_516775 — 2009; 5; contest; pointer; S=0.
- chn_2009_bd0abf — 2009; 3; contest; pointer; S=0.
- chn_2010_7c340a — 2010; 4; contest; pointer; S=0.
- chn_2010_9c2580 — 2010; 5; contest; pointer; S=0.
- chn_2011_a4e35f — 2011; 4; contest; pointer; S=0.
- chn_2011_fe2d36 — 2011; 3; contest; pointer; S=0.
- chn_2012_5657bf — 2012; 2; contest; pointer; S=0.
- chn_2012_6eb96b — 2012; 6; contest; pointer; S=0.
- chn_2012_e3c913 — 2012; 3; contest; pointer; S=0.
- chn_2013_260364 — 2013; 3; contest; pointer; S=0.
- chn_2013_6a3488 — 2013; 4; contest; pointer; S=0.
- chn_2013_895030 — 2013; 5; contest; pointer; S=0.
- chn_2013_912ba5 — 2013; 6; contest; pointer; S=0.
- chn_2013_a3292d — 2013; 3; contest; pointer; S=0.
- chn_2013_b7b259 — 2013; 5; contest; pointer; S=0.
- chn_2022_4f6805 — 2022; 6; contest; text; S=1.
- chn_2022_aece1e — 2022; 3; contest; text; S=0.
- chn_2023_0ba7d6 — 2023; 4; contest; text; S=1.
- chn_2023_422c39 — 2023; 6; contest; text; S=1.
- chn_2023_b3eef6 — 2023; 1; contest; text; S=4.
- chn_2023_ee4f6e — 2023; 3; contest; text; S=2.
- chn_2024_465231 — 2024; 5; contest; text; S=1.
- chn_2024_86460e — 2024; 3; contest; text; S=1.
- chn_2024_9e1916 — 2024; 1; contest; text; S=1.

China-TST

- chn_2007_1e57b6 — 2007; 5.; selection; pointer; S=0.
- chn_2007_976697 — 2007; ?; selection; pointer; S=0.
- chn_2007_a67dc1 — 2007; 3; selection; pointer; S=0.
- chn_2008_e373a0 — 2008; 3; selection; pointer; S=0.
- chn_2009_84a595 — 2009; 5; selection; pointer; S=0.

- chn_2009_caa35c — 2009; 6; selection; pointer; S=0.
- chn_2010_66974c — 2010; 3; selection; pointer; S=0.
- chn_2010_6949ad — 2010; 4; selection; pointer; S=0.
- chn_2010_ad1a21 — 2010; 6; selection; pointer; S=0.
- chn_2011_0b2339 — 2011; 5; selection; pointer; S=0.
- chn_2011_0fc49d — 2011; 3; selection; pointer; S=0.
- chn_2011_110000 — 2011; 6; selection; pointer; S=0.
- chn_2011_5a9253 — 2011; 2; selection; pointer; S=0.
- chn_2011_6cb2c4 — 2011; 1; selection; pointer; S=0.
- chn_2013_0bc239 — 2013; 6; selection; pointer; S=0.
- chn_2013_337c19 — 2013; 3; selection; pointer; S=0.
- chn_2013_694502 — 2013; 4; selection; pointer; S=0.
- chn_2014_1d410b — 2014; 2; selection; text; S=1.
- chn_2021_28dfb3 — 2021; 5; selection; text; S=1.
- chn_2021_36327e — 2021; 3; selection; text; S=1.
- chn_2021_376391 — 2021; 1; selection; text; S=2.
- chn_2021_4e9786 — 2021; 1; selection; text; S=1.
- chn_2021_584851 — 2021; 1; selection; text; S=2.
- chn_2021_8156cd — 2021; 4; selection; text; S=1.
- chn_2021_85d786 — 2021; 2; selection; text; S=1.
- chn_2021_a11094 — 2021; 2; selection; text; S=1.
- chn_2021_ab96ca — 2021; 3; selection; text; S=1.
- chn_2021_cb4abc — 2021; 6; selection; text; S=1.
- chn_2021_d4b623 — 2021; 5; selection; text; S=1.
- chn_2021_e6f1e7 — 2021; 2; selection; text; S=1.
- chn_2021_fd4121 — 2021; 5; selection; text; S=1.
- chn_2022_01f5d7 — 2022; 1; selection; text; S=1.
- chn_2022_1913d8 — 2022; 5; selection; text; S=1.
- chn_2022_43fc7f — 2022; 6; selection; text; S=1.
- chn_2022_4d5044 — 2022; 2; selection; text; S=2.
- chn_2022_514503 — 2022; 3; selection; text; S=1.
- chn_2022_588922 — 2022; 3; selection; text; S=2.
- chn_2022_f8505c — 2022; 6; selection; text; S=1.
- chn_2023_56850c — 2023; 5; selection; text; S=1.
- chn_2023_8e18d4 — 2023; 1; selection; text; S=1.
- chn_2023_a44cf5 — 2023; 6; selection; text; S=1; checked W18.
- chn_2023_e1b280 — 2023; 6; selection; text; S=0.
- chn_2023_f22969 — 2023; 5; selection; text; S=1.
- chn_2025_0db89f — 2025; Problem 6; selection; text; S=1.
- chn_2025_2add97 — 2025; Problem 5; selection; text; S=1.
- chn_2025_53da42 — 2025; Problem 9; selection; text; S=0.
- chn_2025_584da0 — 2025; Problem 3; selection; text; S=1.
- chn_2025_d8572e — 2025; Problem 10; selection; text; S=1.

EGMO

- egm_2012_2d1fb4 — 2012; Problem 2.; contest; text; S=1.
- egm_2012_5667e3 — 2012; Problem 6; contest; text; S=2.
- egm_2012_684025 — 2012; Problem 8; contest; text; S=3.

- egm_2012_b764d7 — 2012; Problem 4.; contest; text; S=1.
- rou_2012_cf5e45 — 2012; Problema 2; contest; text; S=0.
- rou_2012_e2ccd4 — 2012; Problema 8; contest; text; S=0.
- rou_2012_f18db0 — 2012; Problema 4; contest; text; S=0.
- rou_2012_f9e615 — 2012; Problema 6; contest; text; S=0.
- egm_2013_35d4e6 — 2013; Problem 2.; contest; text; S=0.
- egm_2013_6d183f — 2013; Problem 3.; contest; text; S=1.
- egm_2013_a2b3b6 — 2013; Problem 6.; contest; text; S=2.
- egm_2014_0cc86d — 2014; 4.; contest; text; S=1.
- egm_2014_8fdc8e — 2014; 5.; contest; text; S=2.
- mkd_2015_b52719 — 2015; 2; contest; text; S=1.
- mkd_2015_cdfbb6 — 2015; 5; contest; text; S=1.
- mkd_2015_de03cc — 2015; 3; contest; text; S=3.
- rou_2016_9a4499 — 2016; Problem 3.; contest; text; S=1.
- rou_2016_9aa109 — 2016; Problem 5.; contest; text; S=1; checked W24.
- egm_2018_c41bdb — 2018; Problem 3; contest; text; S=2.
- egm_2020_d0be5e — 2020; Problem 4; contest; text; S=4.
- egm_2023_3c6e1c — 2023; Problem 3; contest; text; S=1.
- egm_2023_fc0b21 — 2023; Problem 5; contest; text; S=4.
- egm_2024_10a8ce — 2024; Problem 4; contest; text; S=1.
- egm_2024_86f588 — 2024; Problem 1; contest; text; S=3.
- egm_2025_23b350 — 2025; P2.; contest; text; S=4.
- egm_2025_af59a0 — 2025; P6.; contest; text; S=1.
- egm_2025_b50ee2 — 2025; P5.; contest; text; S=0.

EGMO-TST

- ind_2022_30b7b3 — 2022; Problem 3.; quarantine; pointer; S=0.
- ind_2022_861393 — 2022; 2; quarantine; pointer; S=0.
- ind_2022_ab55c3 — 2022; Problem 1.; quarantine; pointer; S=0.
- ind_2023_88afaf — 2023; Problem 2; quarantine; pointer; S=0.
- ind_2023_960c35 — 2023; 1; quarantine; pointer; S=0.
- ind_2023_9628c9 — 2023; 3; quarantine; pointer; S=0.
- ind_2023_edfa2e — 2023; 2; quarantine; pointer; S=0.
- ind_2024_24a502 — 2024; 2.; quarantine; pointer; S=0.
- ind_2024_bf04ec — 2024; 3; quarantine; pointer; S=0.

IMO

- esp_1983_4d889d — 1983; 5.; contest; text; S=0.
- esp_1983_f5c138 — 1983; 4.; contest; text; S=0.
- esp_1984_25d462 — 1984; 3.; contest; text; S=0.
- esp_1984_28162f — 1984; 5.; contest; text; S=0.
- esp_1985_31cbd9 — 1985; 2.; contest; text; S=0.
- esp_1985_417bfe — 1985; 4.; contest; text; S=0.
- esp_1985_e6ffa9 — 1985; 3.; contest; text; S=0.
- esp_1986_636219 — 1986; 6.; contest; text; S=0.
- esp_1986_ae2a34 — 1986; 3.; contest; text; S=0.
- esp_1987_575956 — 1987; 1.; contest; text; S=0.

- esp_1987_ac6655 — 1987; 3.; contest; text; S=0.
- chn_2003_d213a9 — 2003; 1; contest; pointer; S=0.
- chn_2004_708954 — 2004; 3; contest; pointer; S=0.
- chn_2004_be8846 — 2004; 6; contest; pointer; S=0.
- usa_2005_9b3d8f — 2005; 2; contest; text; S=1.
- usa_2006_d819ea — 2006; 2; contest; text; S=1.
- chn_2007_0e3d67 — 2007; 3; contest; pointer; S=0.
- chn_2008_bad293 — 2008; 5.; contest; pointer; S=0.
- usa_2009_171772 — 2009; 6; contest; text; S=1.
- usa_2009_ed888c — 2009; 5; contest; text; S=2.
- zaf_2012_18a50f — 2012; 6; contest; pointer; S=0.
- zaf_2012_b81c59 — 2012; 3; contest; pointer; S=0.
- esp_2013_8adf5f — 2013; 6; contest; text; S=1.
- esp_2013_93e99b — 2013; 2; contest; text; S=1.
- esp_2014_460932 — 2014; 1; contest; text; S=1.
- esp_2014_b2ff75 — 2014; 5; contest; text; S=1.
- esp_2014_c58f52 — 2014; 6; contest; text; S=1.
- esp_2016_6b7306 — 2016; 2; contest; text; S=0.
- esp_2016_70968a — 2016; 6; contest; text; S=0.

IMO-SL

- imo_2006_2a7e32 — 2006; N7; shortlist; text; S=1.
- imo_2006_47470d — 2006; C7; shortlist; text; S=0.
- imo_2006_816c88 — 2006; C2; shortlist; text; S=2.
- imo_2006_a3af5a — 2006; C4; shortlist; text; S=1.
- imo_2006_b88369 — 2006; C5; shortlist; text; S=1; checked W08.
- imo_2006_dccdc3 — 2006; C1; shortlist; text; S=1.
- imo_2006_f50971 — 2006; C3; shortlist; text; S=2; checked W07.
- imo_2006_fa06e0 — 2006; C6; shortlist; text; S=0.
- imo_2007_09542d — 2007; C1; shortlist; text; S=2.
- imo_2007_104cb8 — 2007; C3; shortlist; text; S=1.
- imo_2007_16b745 — 2007; N1; shortlist; text; S=1.
- imo_2007_2b3b32 — 2007; N3; shortlist; text; S=1.
- imo_2007_524a29 — 2007; A1; shortlist; text; S=1.
- imo_2007_58d0fc — 2007; C2; shortlist; text; S=0.
- imo_2007_67267d — 2007; C7; shortlist; text; S=1; checked W12.
- imo_2007_be294f — 2007; C5; shortlist; text; S=0.
- imo_2007_c1c48b — 2007; C8; shortlist; text; S=0.
- imo_2007_d1b775 — 2007; A2; shortlist; text; S=1.
- imo_2007_e1c86b — 2007; N7; shortlist; text; S=2.
- imo_2007_e391ee — 2007; C4; shortlist; text; S=1.
- imo_2007_f1ecad — 2007; N4; shortlist; text; S=1.
- imo_2008_06a8f7 — 2008; C2; shortlist; text; S=1.
- imo_2008_384044 — 2008; G5; shortlist; text; S=1.
- imo_2008_5311eb — 2008; N4; shortlist; text; S=2.
- imo_2008_740617 — 2008; C5; shortlist; text; S=0.
- imo_2008_ac4e9a — 2008; A3; shortlist; text; S=1.
- imo_2008_bdc772 — 2008; C3; shortlist; text; S=1.

- imo_2008_edd8bc — 2008; C1; shortlist; text; S=0.
- imo_2009_7c4eea — 2009; C2; shortlist; text; S=1.
- imo_2009_872455 — 2009; C8; shortlist; text; S=3.
- imo_2009_af95e8 — 2009; N1; shortlist; text; S=1.
- imo_2009_c42d2f — 2009; C6; shortlist; text; S=0.
- imo_2009_c91ba6 — 2009; C4; shortlist; text; S=1.
- imo_2009_cf1420 — 2009; N2; shortlist; text; S=1.
- imo_2009_de7766 — 2009; C1; shortlist; text; S=1.
- imo_2009_fa6aa0 — 2009; C5; shortlist; text; S=2.
- imo_2010_237f4c — 2010; A7; shortlist; text; S=2.
- imo_2010_3a2ab0 — 2010; N6; shortlist; text; S=1.
- imo_2010_970d18 — 2010; N5; shortlist; text; S=1.
- imo_2010_abf386 — 2010; A4; shortlist; text; S=2.
- imo_2010_ae862f — 2010; N4; shortlist; text; S=1.
- imo_2010_c0ac83 — 2010; C2; shortlist; text; S=2.
- imo_2010_c12e74 — 2010; C7; shortlist; text; S=2.
- imo_2010_cb1492 — 2010; C4; shortlist; text; S=1.
- imo_2010_f50853 — 2010; C5; shortlist; text; S=2.
- imo_2011_1bfa71 — 2011; C6; shortlist; text; S=1.
- imo_2011_359cb8 — 2011; N7; shortlist; text; S=2.
- imo_2011_3bbde9 — 2011; C7; shortlist; text; S=1.
- imo_2011_5fba9c — 2011; A2; shortlist; text; S=1.
- imo_2011_bc8c85 — 2011; C5; shortlist; text; S=0.
- imo_2011_d1d467 — 2011; A4; shortlist; text; S=1.
- imo_2011_f26960 — 2011; N3; shortlist; text; S=1.
- imo_2012_16a6fb — 2012; C3; shortlist; text; S=1.
- imo_2012_21355f — 2012; C1; shortlist; text; S=3.
- imo_2012_2dedb8 — 2012; C2; shortlist; text; S=1.
- imo_2012_35e1cc — 2012; N8; shortlist; text; S=2.
- imo_2012_6b2e95 — 2012; C4; shortlist; text; S=1.
- imo_2012_ddfd08 — 2012; C5; shortlist; text; S=1.
- imo_2013_34622f — 2013; C5; shortlist; text; S=1.
- imo_2013_34c239 — 2013; A2; shortlist; text; S=1.
- imo_2013_3eed83 — 2013; A4; shortlist; text; S=2.
- imo_2013_4d15d8 — 2013; C1; shortlist; text; S=0.
- imo_2013_5a8513 — 2013; C7; shortlist; text; S=0.
- imo_2013_5ee60e — 2013; C8; shortlist; text; S=1.
- imo_2013_7500b7 — 2013; C3; shortlist; text; S=2.
- imo_2013_a66da8 — 2013; N7; shortlist; text; S=1.
- imo_2013_ca3c20 — 2013; A5; shortlist; text; S=1.
- imo_2013_d960d1 — 2013; A1; shortlist; text; S=1.
- imo_2013_eb8b9e — 2013; C2; shortlist; text; S=2.
- imo_2013_f9ba1b — 2013; C4; shortlist; text; S=1.
- imo_2014_013ae9 — 2014; C6; shortlist; text; S=1.
- imo_2014_09db9c — 2014; C8; shortlist; text; S=1.
- imo_2014_44369f — 2014; C3; shortlist; text; S=0.
- imo_2014_4fbd86 — 2014; A3; shortlist; text; S=1.
- imo_2014_6534d6 — 2014; A1; shortlist; text; S=1.
- imo_2014_d42980 — 2014; C5; shortlist; text; S=0.

- imo_2014_da4751 — 2014; N6; shortlist; text; S=2.
- imo_2014_fb1712 — 2014; C7; shortlist; text; S=0.
- imo_2015_0681fc — 2015; C1; shortlist; text; S=3.
- imo_2015_0b0e39 — 2015; A3; shortlist; text; S=2.
- imo_2015_207240 — 2015; N6; shortlist; text; S=1.
- imo_2015_552a15 — 2015; C7; shortlist; text; S=0.
- imo_2015_8f5283 — 2015; A6; shortlist; text; S=2.
- imo_2015_ae4598 — 2015; C2; shortlist; text; S=1.
- imo_2015_bab8ff — 2015; C5; shortlist; text; S=1.
- imo_2015_f31319 — 2015; C6; shortlist; text; S=2.
- imo_2016_5fca7b — 2016; C6; shortlist; text; S=1.
- imo_2016_88479f — 2016; C4; shortlist; text; S=1.
- imo_2016_b6b418 — 2016; A3; shortlist; text; S=2.
- imo_2016_f01bd2 — 2016; C7; shortlist; text; S=0.
- imo_2016_f9df36 — 2016; C2; shortlist; text; S=1.
- imo_2016_fa5aef — 2016; N8; shortlist; text; S=1.
- imo_2017_2878d3 — 2017; N2; shortlist; text; S=1.
- imo_2017_38c1ed — 2017; A5; shortlist; text; S=1.
- imo_2017_403e3f — 2017; C7; shortlist; text; S=2.
- imo_2017_7f8a8b — 2017; C2; shortlist; text; S=2.
- imo_2017_a6d9fd — 2017; G8; shortlist; text; S=0.
- imo_2017_abff99 — 2017; C8; shortlist; text; S=0.
- imo_2017_de14b7 — 2017; C3; shortlist; text; S=1.
- imo_2017_f79d3e — 2017; N7; shortlist; text; S=2.
- imo_2018_11785f — 2018; N7; shortlist; text; S=1.
- imo_2018_174f68 — 2018; N6; shortlist; text; S=2.
- imo_2018_55614a — 2018; C4; shortlist; text; S=0; checked W15.
- imo_2018_556799 — 2018; C6; shortlist; text; S=2.
- imo_2018_b7dcfb — 2018; C5; shortlist; text; S=2.
- imo_2018_ee943a — 2018; C7; shortlist; text; S=2.
- imo_2018_eed763 — 2018; C2; shortlist; text; S=0; checked W16.
- imo_2018_f30e16 — 2018; A6; shortlist; text; S=1.
- imo_2019_05dd86 — 2019; N3; shortlist; text; S=2.
- imo_2019_070c84 — 2019; A3; shortlist; text; S=4.
- imo_2019_152467 — 2019; N7; shortlist; text; S=1.
- imo_2019_26d060 — 2019; C3; shortlist; text; S=4.
- imo_2019_335dd1 — 2019; C4; shortlist; text; S=1.
- imo_2019_486790 — 2019; C8; shortlist; text; S=1.
- imo_2019_51a004 — 2019; C7; shortlist; text; S=1.
- imo_2019_7bd8a9 — 2019; G8; shortlist; text; S=2.
- imo_2019_abb25c — 2019; A1; shortlist; text; S=2.
- imo_2019_aedde5 — 2019; N5; shortlist; text; S=2.
- imo_2019_c92128 — 2019; C5; shortlist; text; S=0.
- imo_2019_ce6a78 — 2019; N6; shortlist; text; S=4.
- imo_2019_e753a3 — 2019; C2; shortlist; text; S=2.
- imo_2019_f43f3b — 2019; C1; shortlist; text; S=1.
- imo_2019_fa57ae — 2019; C9; shortlist; text; S=1.
- imo_2020_19e099 — 2020; C1; shortlist; text; S=2.
- imo_2020_48d762 — 2020; A1; shortlist; text; S=1.

- imo_2020_9168c0 — 2020; C7; shortlist; text; S=1.
- imo_2020_c1c2cc — 2020; C2; shortlist; text; S=1.
- imo_2020_e10061 — 2020; C6; shortlist; text; S=2.
- imo_2021_1c3cfc — 2021; N4; shortlist; text; S=1.
- imo_2021_4e49c6 — 2021; C8; shortlist; text; S=2.
- imo_2021_5bfb5f — 2021; C1; shortlist; text; S=1.
- imo_2021_862aab — 2021; N6; shortlist; text; S=1.
- imo_2021_a06079 — 2021; C3; shortlist; text; S=1.
- imo_2021_a8deb3 — 2021; C4; shortlist; text; S=2.
- imo_2021_b0420a — 2021; C6; shortlist; text; S=1.
- imo_2021_da9f3d — 2021; C2; shortlist; text; S=1.
- imo_2021_f2791d — 2021; A3; shortlist; text; S=3.
- imo_2021_fbb8aa — 2021; C7; shortlist; text; S=0.
- imo_2022_1937c6 — 2022; A8.; shortlist; text; S=1.
- imo_2022_264ed6 — 2022; N8.; shortlist; text; S=2.
- imo_2022_34151b — 2022; C8.; shortlist; text; S=0.
- imo_2022_4008eb — 2022; C7.; shortlist; text; S=1.
- imo_2022_57242c — 2022; A2.; shortlist; text; S=1.
- imo_2022_70d092 — 2022; C9.; shortlist; text; S=1.
- imo_2022_783196 — 2022; C6.; shortlist; text; S=3.
- imo_2022_b034b8 — 2022; C2.; shortlist; text; S=1.
- imo_2022_d417bb — 2022; N5.; shortlist; text; S=1.
- imo_2022_e5b1cb — 2022; C1.; shortlist; text; S=1.
- imo_2023_375cf9 — 2023; C4.; shortlist; text; S=3.
- imo_2023_73cc8d — 2023; C7.; shortlist; text; S=1.
- imo_2023_76bf3b — 2023; A5.; shortlist; text; S=1.
- imo_2023_7da218 — 2023; C6.; shortlist; pointer; S=0.
- imo_2023_913777 — 2023; C5.; shortlist; text; S=2.
- imo_2023_a99802 — 2023; C3.; shortlist; pointer; S=0.
- imo_2023_b2d73c — 2023; A1.; shortlist; text; S=2.
- imo_2023_d4d6f3 — 2023; A6.; shortlist; text; S=1.
- imo_2023_f7b402 — 2023; A7.; shortlist; text; S=2.
- imo_2024_0968d7 — 2024; C2; shortlist; text; S=1.
- imo_2024_1fd9d0 — 2024; C5; shortlist; text; S=1.
- imo_2024_240890 — 2024; A6; shortlist; text; S=1.
- imo_2024_34fef6 — 2024; C8; shortlist; text; S=0.
- imo_2024_3f4e9a — 2024; N6; shortlist; text; S=4.
- imo_2024_5877d2 — 2024; A2; shortlist; text; S=2.
- imo_2024_657c6a — 2024; C1; shortlist; text; S=4.
- imo_2024_7201b9 — 2024; N7; shortlist; text; S=2.
- imo_2024_7579e8 — 2024; C4; shortlist; text; S=0.
- imo_2024_91fd1f — 2024; C6; shortlist; text; S=1.
- imo_2024_95e706 — 2024; C7; shortlist; text; S=2.

IMO-TST

- usa_87d02d — ?; Problem 2; selection; text; S=1.
- usa_2001_419b80 — 2001; 9; selection; text; S=1.
- usa_2001_77e185 — 2001; 4; selection; text; S=1.

- usa_2001_8a88f8 — 2001; 3; selection; text; S=1.
- usa_2001_9b2113 — 2001; 4; selection; text; S=1.
- usa_2001_a0d6f2 — 2001; 2; selection; text; S=1.
- usa_2001_e83e65 — 2001; 3; selection; text; S=2.
- vnm_2001_603cc9 — 2001; 3; selection; text; S=0.
- vnm_2001_644f7b — 2001; 5; selection; text; S=0.
- vnm_2001_95fe00 — 2001; 6; selection; text; S=0.
- vnm_2002_aea91a — 2002; Problem 4; selection; text; S=0.
- vnm_2002_ba609d — 2002; Problem 2; selection; text; S=0.
- usa_2003_73c7f7 — 2003; 1; selection; pointer; S=0.
- vnm_2003_23ab5b — 2003; 3; selection; text; S=0.
- vnm_2003_5c9cad — 2003; 1; selection; text; S=0.
- vnm_2003_767bbb — 2003; 5; selection; text; S=0.
- usa_2004_8f5af1 — 2004; 4; selection; text; S=0.
- usa_2004_954202 — 2004; 3; selection; text; S=1.
- usa_2004_cbf0f1 — 2004; 2; selection; text; S=1.
- usa_2006_6ff7ae — 2006; 1; selection; text; S=0.
- usa_2006_b160c2 — 2006; 5; selection; text; S=0.
- usa_2007_c129df — 2007; 6; selection; text; S=1.
- usa_2007_f12e75 — 2007; 2; selection; text; S=1.
- usa_2008_4ddab6 — 2008; 1; selection; text; S=1.
- usa_2008_cbf07d — 2008; 5; selection; text; S=2.
- usa_2008_de1133 — 2008; 3; selection; text; S=1.
- usa_2009_3358a9 — 2009; 3; selection; text; S=1.
- usa_2009_ef3309 — 2009; 6; selection; text; S=2.
- usa_2009_f1ac89 — 2009; 1; selection; text; S=2.
- ind_2010_051b62 — 2010; 5; selection; text; S=1.
- sgp_2010_5727a3 — 2010; 10; selection; text; S=1.
- sgp_2010_6c0264 — 2010; 6; selection; text; S=1.
- sgp_2010_7e3a81 — 2010; 8; selection; text; S=1.
- sgp_2010_c6cc18 — 2010; 2; selection; text; S=1.
- usa_2010_061aff — 2010; 6; selection; text; S=1.
- usa_2010_773ef5 — 2010; 8; selection; text; S=2.
- usa_2010_7fb109 — 2010; 5; selection; text; S=1.
- vnm_2010_5c6598 — 2010; Problem 5; selection; text; S=0.
- vnm_2010_6c1af7 — 2010; Problem 3; selection; pointer; S=0.
- ind_2011_9fc5bb — 2011; 4; selection; text; S=1.
- usa_2011_02c62b — 2011; 5; selection; text; S=1.
- usa_2011_5c5516 — 2011; 8; selection; text; S=1.
- usa_2011_8b387a — 2011; 3; selection; text; S=1.
- usa_2011_c4196c — 2011; 9; selection; text; S=1.
- usa_2011_ff9f97 — 2011; 6; selection; text; S=1.
- vnm_2011_c8e35c — 2011; 6; selection; text; S=0.
- vnm_2011_f6e2ec — 2011; 1; selection; text; S=0.
- ind_2012_2ce73f — 2012; 8; selection; text; S=1.
- ind_2012_f07169 — 2012; 9; selection; text; S=1.
- usa_2012_3b4732 — 2012; 5; selection; text; S=1.
- usa_2012_485e07 — 2012; 3; selection; text; S=1.
- usa_2012_5521cf — 2012; 4; selection; text; S=1.

- usa_2012_5abc60 — 2012; 7; selection; text; S=1.
- usa_2012_af9e07 — 2012; 9.; selection; text; S=1.
- usa_2012_e9f6ee — 2012; 3.; selection; text; S=1.
- vnm_2012_1c51a9 — 2012; B?i 6; selection; text; S=0.
- vnm_2012_6d51b3 — 2012; B?i 2; selection; text; S=0.
- vnm_2012_f548c9 — 2012; B?i 3; selection; text; S=0.
- ind_2013_5ef73e — 2013; 3; selection; text; S=1; checked W20.
- ind_2013_abf1b9 — 2013; 4; selection; text; S=1.
- usa_2013_2d2e1e — 2013; 1.; selection; text; S=1.
- usa_2013_536721 — 2013; 7.; selection; text; S=1.
- usa_2013_7b965d — 2013; 8.; selection; pointer; S=1.
- usa_2013_88fa26 — 2013; 7.; selection; text; S=1.
- usa_2013_bfa49c — 2013; 1.; selection; text; S=1; checked W11.
- usa_2013_f4a0e2 — 2013; 9.; selection; text; S=1.
- usa_2014_402d1a — 2014; 2.; selection; text; S=1.
- usa_2014_c1a41c — 2014; 3.; selection; text; S=1.
- usa_2014_d4dc03 — 2014; 3.; selection; text; S=0.
- usa_2014_e7d2f3 — 2014; 7.; selection; text; S=2.
- vnm_2015_0c8e24 — 2015; 1; selection; text; S=1.
- ind_2016_85f4b8 — 2016; 4; selection; text; S=1.
- ind_2016_9a166a — 2016; 5; selection; text; S=1; checked W05.
- ind_2016_c1a286 — 2016; 2; selection; text; S=1.
- vnm_2017_71cd4a — 2017; Problem 1; selection; text; S=1.
- vnm_2017_d99be9 — 2017; Problem 5; selection; text; S=1.
- vnm_2017_e0f625 — 2017; Problem 6; selection; text; S=1.
- usa_2018_259657 — 2018; TSTST 6; selection; text; S=1.
- usa_2018_5cd6f2 — 2018; TSTST 7; selection; text; S=1.
- usa_2018_7ab8a8 — 2018; TSTST 9; selection; text; S=1.
- usa_2018_9f41d2 — 2018; TSTST 2; selection; text; S=1.
- vnm_2018_027413 — 2018; Problem 5; selection; text; S=0.
- vnm_2018_e63671 — 2018; Problem 2; selection; text; S=0.
- usa_2019_213e85 — 2019; TST 5; selection; text; S=0.
- usa_2019_272f92 — 2019; TST 4; selection; text; S=1.
- usa_2019_2f0721 — 2019; TSTST 3; selection; text; S=0.
- usa_2019_5cb5ea — 2019; TST 3; selection; text; S=0.
- usa_2019_afbc19 — 2019; TSTST 4; selection; text; S=1.
- usa_2019_c48ff2 — 2019; TSTST 6; selection; text; S=1.
- usa_2019_d4cda2 — 2019; TSTST 8; selection; text; S=0.
- usa_2020_4ac916 — 2020; TST 4; selection; text; S=1.
- usa_2020_5d31dd — 2020; TST 1; selection; text; S=1.
- usa_2020_8ea9dc — 2020; TST 3; selection; text; S=0.
- usa_2020_93693e — 2020; TSTST 5; selection; text; S=0.
- usa_2020_b23173 — 2020; TSTST 1; selection; text; S=0.
- usa_2020_fe77d1 — 2020; TSTST 9; selection; text; S=1.
- vnm_2021_ae2abb — 2021; 2; selection; text; S=0.
- vnm_2021_ecd237 — 2021; Problem 6.; selection; text; S=1.
- chn_2022_252cd5 — 2022; 4; selection; text; S=1.
- chn_2022_3e5903 — 2022; 6; selection; text; S=1.
- chn_2022_5a13de — 2022; 5; selection; text; S=1.

- chn_2022_64f1b5 — 2022; 3; selection; text; S=1.
- chn_2022_78f104 — 2022; 5; selection; text; S=1.
- chn_2022_9e340b — 2022; 4; selection; text; S=1.
- chn_2022_bec0c9 — 2022; 3; selection; text; S=1.
- chn_2022_f17616 — 2022; 3; selection; text; S=1.
- vnm_2022_b14250 — 2022; 2; selection; text; S=1.
- vnm_2022_b8b08c — 2022; Problem 6; selection; text; S=1.
- chn_2023_0510b2 — 2023; 5; selection; text; S=1.
- chn_2023_4ceef1 — 2023; 3; selection; text; S=0.
- chn_2023_a2bd30 — 2023; 2; selection; text; S=1.
- chn_2023_b8f301 — 2023; 5; selection; text; S=1.
- chn_2023_d02731 — 2023; 4; selection; text; S=2.
- chn_2023_e54b7e — 2023; 3; selection; text; S=1.
- ind_2023_001b9f — 2023; 1; selection; text; S=1.
- ind_2023_6854af — 2023; Problem 1; selection; text; S=2.
- ind_2023_78c50a — 2023; Problem 3; selection; text; S=1.
- ind_2023_b690b1 — 2023; Problem 2; selection; text; S=3.
- usa_2023_103f73 — 2023; 3; selection; text; S=0.
- usa_2023_4eb572 — 2023; USA TST 2023/3; selection; text; S=1.
- usa_2023_942b5a — 2023; USA TST 2023/1; selection; text; S=0.
- usa_2023_9d8847 — 2023; USA TST 2023/5; selection; text; S=0.
- usa_2023_ca255f — 2023; 2; selection; text; S=1.
- usa_2023_cb6858 — 2023; 9; selection; text; S=1.
- usa_2023_d384cb — 2023; 4; selection; text; S=1.
- usa_2023_de1495 — 2023; USA TST 2023/6; selection; text; S=1.
- usa_2023_e8656c — 2023; 7; selection; text; S=0.
- vnm_2023_3ebb4e — 2023; 1; selection; text; S=1.
- vnm_2023_7638b1 — 2023; 6; selection; text; S=1.
- chn_2024_0b08db — 2024; 1; selection; pointer; S=0.
- chn_2024_2a0411 — 2024; 1; selection; text; S=1.
- chn_2024_420fd0 — 2024; 3; selection; text; S=1.
- chn_2024_44289a — 2024; 3; selection; text; S=1.
- chn_2024_7afa57 — 2024; 6; selection; text; S=2.
- chn_2024_7ed496 — 2024; 4; selection; text; S=1.
- chn_2024_cc0bb9 — 2024; 2; selection; text; S=1.
- ind_2024_0dddaf — 2024; 2.; selection; text; S=2.
- ind_2024_804c67 — 2024; 1; selection; text; S=2.
- ind_2024_af0b04 — 2024; 2.; selection; text; S=1.
- ind_2024_be9bfd — 2024; 3; selection; text; S=1.
- ind_2024_d0a341 — 2024; 2.; selection; text; S=2.
- ind_2024_d6cfca — 2024; 3.; selection; text; S=2.
- vnm_2024_befa2a — 2024; 2; selection; text; S=0.
- chn_2025_2bb7dd — 2025; Problem 1.; selection; text; S=1.
- chn_2025_4d380b — 2025; Problem 10.; selection; text; S=1.
- chn_2025_54b9d7 — 2025; Problem 11.; selection; text; S=0.
- chn_2025_d18489 — 2025; Problem 3.; selection; text; S=1.
- chn_2025_d307ea — 2025; Problem 8.; selection; text; S=1.
- chn_2025_e6abc2 — 2025; Problem 12.; selection; text; S=1.
- chn_2025_f1f576 — 2025; Problem 5.; selection; text; S=1.

- chn_2025_f857cf — 2025; Problem 7.; selection; text; S=1.
- usa_2025_061227 — 2025; TSTST 1; selection; text; S=1.
- usa_2025_3e982f — 2025; Problem 6; selection; text; S=0.
- usa_2025_5085de — 2025; Problem 1; selection; text; S=0.
- usa_2025_c87ac0 — 2025; 1; selection; text; S=0.
- usa_2025_d1cd24 — 2025; TST 1; selection; text; S=1.
- usa_2025_e105bb — 2025; 5; selection; text; S=0.

IMOTC

- ind_2023_092f9b — 2023; Problem 3; pipeline supplement; text; S=1.
- ind_2023_407ed2 — 2023; Problem 1; pipeline supplement; text; S=0.
- ind_2023_d9a3be — 2023; Problem 3; pipeline supplement; text; S=0.

IranMO

- irn_2010_27dfaf — 2010; 8; contest; pointer; S=0.
- irn_2010_3a5950 — 2010; 2; contest; pointer; S=0.
- irn_2010_5f459a — 2010; 8; contest; pointer; S=0.
- irn_2010_7fa7c5 — 2010; 3; contest; pointer; S=0.
- irn_2010_c2ad22 — 2010; 4; contest; pointer; S=0.
- irn_2010_f0a217 — 2010; 6; contest; pointer; S=0.
- irn_2012_1b5675 — 2012; 7; contest; pointer; S=0.
- irn_2012_1d14ab — 2012; 1; contest; pointer; S=0.
- irn_2012_2a6aca — 2012; 12; contest; pointer; S=0.
- irn_2012_31b05c — 2012; 17; contest; pointer; S=0.
- irn_2012_753df8 — 2012; 7; contest; pointer; S=0.
- irn_2012_ceed72 — 2012; 1; contest; pointer; S=0.
- irn_2012_cf8d83 — 2012; 4; contest; pointer; S=0.
- irn_2012_f689b6 — 2012; 5; contest; pointer; S=0.
- irn_2012_fb65c2 — 2012; 5; contest; pointer; S=0.
- irn_2013_5a643b — 2013; 4; contest; pointer; S=0.
- irn_2013_a374f3 — 2013; 7; contest; pointer; S=0.
- irn_2013_bfa84f — 2013; 7; contest; pointer; S=0.
- irn_2013_fde5ec — 2013; 3; contest; pointer; S=0.
- irn_2014_252cc4 — 2014; 5.; contest; pointer; S=0.
- irn_2014_6f786d — 2014; 5.; contest; pointer; S=0.
- irn_2014_6fd317 — 2014; 6.; contest; pointer; S=0.
- irn_2014_75911d — 2014; 9.; contest; pointer; S=0.
- irn_2014_883fdc — 2014; 4.; contest; pointer; S=0.
- irn_2014_8d2bd8 — 2014; 4.; contest; pointer; S=0.
- irn_2014_d056e6 — 2014; 12.; contest; pointer; S=0.
- irn_2014_d49f75 — 2014; 7.; contest; pointer; S=0.
- irn_2015_4e48b5 — 2015; 4; contest; text; S=1.
- irn_2015_5e81d0 — 2015; 9; contest; text; S=0.
- irn_2015_6b4790 — 2015; 3; contest; text; S=1.
- irn_2015_80e073 — 2015; 8; contest; text; S=1.
- irn_2015_8e5525 — 2015; 5; contest; text; S=1.
- irn_2015_acb8a8 — 2015; 1; contest; text; S=1.

- irn_2015_b3bad7 — 2015; 2; contest; pointer; S=1.
- irn_2015_c7ea96 — 2015; 11; contest; text; S=1.
- irn_2015_d475c2 — 2015; 5; contest; pointer; S=1.
- irn_2015_f89b52 — 2015; 5; contest; text; S=1.
- irn_2016_193e9f — 2016; 7.; contest; pointer; S=0.
- irn_2016_308ebc — 2016; 3.; contest; pointer; S=0.
- irn_2016_8173e7 — 2016; 2.; contest; pointer; S=0.
- irn_2016_fd3724 — 2016; 4.; contest; pointer; S=0.
- irn_2017_01b447 — 2017; 16; contest; pointer; S=0.
- irn_2017_2e2167 — 2017; 2; contest; pointer; S=0.
- irn_2017_4e0f36 — 2017; 2; contest; pointer; S=0.
- irn_2017_59fbd9 — 2017; 11; contest; pointer; S=0.
- irn_2017_b68032 — 2017; 9; contest; pointer; S=0.
- irn_2017_f15b37 — 2017; 8; contest; pointer; S=0.
- irn_2017_f908eb — 2017; 3; contest; pointer; S=0.
- irn_2018_12d0a4 — 2018; 8; contest; pointer; S=0.
- irn_2018_398e65 — 2018; 8; contest; pointer; S=0.
- irn_2018_4f15fa — 2018; 6; contest; pointer; S=0.
- irn_2018_63237e — 2018; 12; contest; pointer; S=0.
- irn_2018_816461 — 2018; 9; contest; pointer; S=0.
- irn_2018_9b0ecd — 2018; 4; contest; pointer; S=0.
- irn_2018_fd9cb5 — 2018; 17; contest; pointer; S=0.
- irn_2019_2729ae — 2019; 2; contest; pointer; S=0.
- irn_2019_42df37 — 2019; 2; contest; pointer; S=0.
- irn_2019_43fbcf — 2019; 2; contest; pointer; S=0.
- irn_2019_48349f — 2019; 4; contest; pointer; S=0.
- irn_2019_4a4940 — 2019; 5; contest; pointer; S=0.
- irn_2019_7a8502 — 2019; 4; contest; pointer; S=0.
- irn_2019_7eecf9 — 2019; 17; contest; pointer; S=0.
- irn_2019_877aa4 — 2019; 11; contest; pointer; S=0.
- irn_2019_88c4a6 — 2019; 1; contest; pointer; S=0.
- irn_2019_9018e5 — 2019; 7; contest; pointer; S=0.
- irn_2019_af8b7a — 2019; 1; contest; pointer; S=0.
- irn_2019_c2fee3 — 2019; 1; contest; pointer; S=0.
- irn_2019_ce0ddf — 2019; 10; contest; pointer; S=0.
- irn_2019_ce33fb — 2019; 9; contest; pointer; S=0.
- irn_2019_e7aab5 — 2019; 3; contest; pointer; S=0.
- irn_2020_092093 — 2020; 5; contest; pointer; S=0.
- irn_2020_0bc54b — 2020; 1; contest; pointer; S=0.
- irn_2020_1377d8 — 2020; 2; contest; pointer; S=0.
- irn_2020_1f2c39 — 2020; 5; contest; pointer; S=0.
- irn_2020_20e479 — 2020; 7; contest; pointer; S=0.
- irn_2020_214101 — 2020; 8; contest; pointer; S=0.
- irn_2020_36e84a — 2020; 5; contest; pointer; S=0.
- irn_2020_4f9267 — 2020; 1; contest; pointer; S=0.
- irn_2020_6b1d80 — 2020; 6; contest; pointer; S=0.
- irn_2020_944c87 — 2020; 11; contest; pointer; S=0.
- irn_2020_969ea4 — 2020; 5; contest; pointer; S=0.
- irn_2020_d88957 — 2020; 6; contest; pointer; S=0.

- irn_2020_decf38 — 2020; 12; contest; pointer; S=0.
- irn_2020_fc7c20 — 2020; 3; contest; pointer; S=0.
- irn_2021_102b67 — 2021; 12; contest; pointer; S=0.
- irn_2021_115e75 — 2021; 7; contest; pointer; S=0.
- irn_2021_415869 — 2021; 5; contest; pointer; S=0.
- irn_2021_45847b — 2021; 4; contest; pointer; S=0.
- irn_2021_578e8a — 2021; 2; contest; pointer; S=0.
- irn_2021_61655e — 2021; 3; contest; pointer; S=0.
- irn_2021_72be43 — 2021; 6; contest; pointer; S=0.
- irn_2021_831c1f — 2021; 2; contest; pointer; S=0.
- irn_2021_869f28 — 2021; 1; contest; pointer; S=0.
- irn_2021_a0ce3a — 2021; 1; contest; pointer; S=0.
- irn_2021_afea16 — 2021; 4; contest; pointer; S=0.
- irn_2021_cbdef8 — 2021; 5; contest; pointer; S=0.
- irn_2022_1e167e — 2022; 4; contest; pointer; S=0.
- irn_2022_20fab9 — 2022; 6; contest; pointer; S=0.
- irn_2022_210321 — 2022; 1; contest; pointer; S=0.
- irn_2022_33d6c2 — 2022; 5; contest; pointer; S=0.
- irn_2022_ae1388 — 2022; 2; contest; pointer; S=0.
- irn_2022_c2ed96 — 2022; 6; contest; pointer; S=0.
- irn_2022_d6a5b8 — 2022; 1; contest; pointer; S=0.
- irn_2022_d75975 — 2022; 1; contest; pointer; S=0.
- irn_2022_e970d2 — 2022; 9; contest; pointer; S=0.
- irn_2023_0c7844 — 2023; 1; contest; pointer; S=0.
- irn_2023_146bc7 — 2023; 1; contest; pointer; S=0.
- irn_2023_2a4c72 — 2023; 6; contest; pointer; S=0.
- irn_2023_38714a — 2023; 2; contest; pointer; S=0.
- irn_2023_38bb7e — 2023; 2; contest; pointer; S=0.
- irn_2023_484855 — 2023; 4; contest; pointer; S=0.
- irn_2023_6607af — 2023; 5; contest; pointer; S=0.
- irn_2023_66bca9 — 2023; 3; contest; pointer; S=0.
- irn_2023_74f4b3 — 2023; 3; contest; pointer; S=0.
- irn_2023_9013f9 — 2023; 4; contest; pointer; S=0.
- irn_2023_a1ad16 — 2023; 3; contest; pointer; S=0.
- irn_2023_a34cad — 2023; 1; contest; pointer; S=0.
- irn_2023_c6c5ca — 2023; 5; contest; pointer; S=0.
- irn_2023_d16e44 — 2023; 3; contest; pointer; S=0.
- irn_2023_d228c3 — 2023; 4; contest; pointer; S=0.
- irn_2023_d7aef1 — 2023; 6; contest; pointer; S=0.
- irn_2024_046b9f — 2024; 4; contest; pointer; S=0.
- irn_2024_1a326f — 2024; 4; contest; pointer; S=0.
- irn_2024_2c2f04 — 2024; 6; contest; pointer; S=0.
- irn_2024_43e995 — 2024; 1; contest; pointer; S=0.
- irn_2024_9616c3 — 2024; 1; contest; pointer; S=0.
- irn_2024_9bb217 — 2024; 2; contest; pointer; S=0.
- irn_2024_a04e9a — 2024; 5; contest; pointer; S=0.
- irn_2024_ba40e0 — 2024; 5; contest; pointer; S=0.
- irn_2024_f206ba — 2024; 3; contest; pointer; S=0.
- irn_2025_033975 — 2025; 5; contest; pointer; S=0.

- irn_2025_300b39 — 2025; 4; contest; pointer; S=0.
- irn_2025_3bae2c — 2025; 2; contest; pointer; S=0.
- irn_2025_5b51d8 — 2025; 1; contest; pointer; S=0.
- irn_2025_664197 — 2025; 2; contest; pointer; S=0.
- irn_2025_916f58 — 2025; 3; contest; pointer; S=0.
- irn_2025_9c031b — 2025; 2; contest; pointer; S=0.
- irn_2025_b8fd6f — 2025; 1; contest; pointer; S=0.
- irn_2025_c9c17e — 2025; 3; contest; pointer; S=0.
- irn_2025_dace5a — 2025; 1; contest; pointer; S=0.
- irn_2025_fbd658 — 2025; 5; contest; pointer; S=0.

JMO

- jpn_2006_1f6f07 — 2006; 6; pipeline supplement; text; S=1.
- jpn_2006_2aa164 — 2006; 12; pipeline supplement; text; S=1.
- jpn_2006_9417a1 — 2006; 4; pipeline supplement; text; S=0.
- jpn_2006_a53c60 — 2006; 8; pipeline supplement; text; S=1.
- jpn_2006_ef0ba3 — 2006; 3; pipeline supplement; text; S=1.
- jpn_2006_ff34be — 2006; 10; pipeline supplement; text; S=0.
- jpn_2007_07e0bb — 2007; 5; pipeline supplement; text; S=1.
- jpn_2007_2549e4 — 2007; 4; pipeline supplement; text; S=0.
- jpn_2007_259143 — 2007; 12; pipeline supplement; text; S=1.
- jpn_2007_406148 — 2007; 9; pipeline supplement; text; S=0.
- jpn_2007_44d09d — 2007; 8; pipeline supplement; text; S=1.
- jpn_2007_6677e6 — 2007; 1; pipeline supplement; text; S=1.
- jpn_2007_7ed5b8 — 2007; 7; pipeline supplement; text; S=1.
- jpn_2007_96ded9 — 2007; 12; pipeline supplement; text; S=1.
- jpn_2007_9c69b3 — 2007; 5; pipeline supplement; text; S=1.
- jpn_2007_a14cf1 — 2007; 10; pipeline supplement; text; S=1.
- jpn_2007_c7dc24 — 2007; 1; pipeline supplement; text; S=1.
- jpn_2007_ed31dc — 2007; 5; pipeline supplement; text; S=1.
- jpn_2007_f72375 — 2007; 6; pipeline supplement; text; S=1.
- jpn_2008_1c7a4f — 2008; 8; pipeline supplement; text; S=1.
- jpn_2008_7fbffd — 2008; 10; pipeline supplement; text; S=1.
- jpn_2008_945846 — 2008; 5; pipeline supplement; text; S=1.
- jpn_2008_ade892 — 2008; 12; pipeline supplement; text; S=1.
- jpn_2008_cd2379 — 2008; 2; pipeline supplement; text; S=1.
- jpn_2008_eec352 — 2008; 3; pipeline supplement; text; S=1.
- jpn_2009_0a1555 — 2009; 12; pipeline supplement; text; S=1.
- jpn_2009_4f9146 — 2009; 9; pipeline supplement; text; S=1.
- jpn_2009_52732e — 2009; 5; pipeline supplement; text; S=1.
- jpn_2009_981d39 — 2009; 5; pipeline supplement; text; S=1.
- jpn_2009_c83064 — 2009; 2; pipeline supplement; text; S=2.
- jpn_2010_04151c — 2010; 5; pipeline supplement; text; S=1.
- jpn_2010_1e8d8f — 2010; 1; pipeline supplement; text; S=1.
- jpn_2010_2a8898 — 2010; 10; pipeline supplement; text; S=1.
- jpn_2010_3b3a8d — 2010; 2; pipeline supplement; text; S=1.
- jpn_2010_3dd59d — 2010; 5; pipeline supplement; text; S=1.
- jpn_2010_3f9b46 — 2010; 7; pipeline supplement; text; S=1.

- jpn_2010_5ba937 — 2010; 3; pipeline supplement; text; S=2.
- jpn_2010_61aff6 — 2010; 12; pipeline supplement; text; S=1.
- jpn_2010_869fd1 — 2010; 6; pipeline supplement; text; S=1.
- jpn_2010_b192db — 2010; 2; pipeline supplement; text; S=1.
- jpn_2010_c054ab — 2010; 9; pipeline supplement; text; S=1.
- jpn_2011_570513 — 2011; 7; pipeline supplement; text; S=1.
- jpn_2011_7b87b7 — 2011; 3; pipeline supplement; text; S=1.
- jpn_2011_8c8ec4 — 2011; 9; pipeline supplement; text; S=1.
- jpn_2011_9ace44 — 2011; 1; pipeline supplement; text; S=1.
- jpn_2012_365658 — 2012; 3; pipeline supplement; text; S=1.
- jpn_2012_592bd2 — 2012; 6; pipeline supplement; text; S=0.
- jpn_2012_70a429 — 2012; 8; pipeline supplement; text; S=1.
- jpn_2012_8619cd — 2012; 11; pipeline supplement; text; S=1.
- jpn_2012_993c1a — 2012; 5; pipeline supplement; text; S=1.
- jpn_2012_b3b037 — 2012; 5; pipeline supplement; text; S=0.
- jpn_2013_050803 — 2013; 7; pipeline supplement; text; S=0.
- jpn_2013_348c1d — 2013; 5; pipeline supplement; text; S=0.
- jpn_2013_45e694 — 2013; 1; pipeline supplement; text; S=1.
- jpn_2013_519e11 — 2013; 11; pipeline supplement; text; S=0.
- jpn_2013_6a6bc7 — 2013; 5; pipeline supplement; text; S=0.
- jpn_2013_798f65 — 2013; 10; pipeline supplement; text; S=1.
- jpn_2013_c38542 — 2013; 12; pipeline supplement; text; S=1.
- jpn_2013_eb415f — 2013; 6; pipeline supplement; text; S=1.
- jpn_2014_69b91c — 2014; 7; pipeline supplement; text; S=0.
- jpn_2014_74269a — 2014; 6; pipeline supplement; text; S=1.
- jpn_2014_8f91d2 — 2014; 2; pipeline supplement; text; S=0.
- jpn_2014_a052c4 — 2014; 5; pipeline supplement; text; S=1.
- jpn_2014_adb623 — 2014; 3; pipeline supplement; text; S=1.
- jpn_2014_bd1bef — 2014; 11; pipeline supplement; text; S=1.
- jpn_2014_f4f5dc — 2014; 10; pipeline supplement; text; S=1.
- jpn_2015_2c74e3 — 2015; 5; pipeline supplement; text; S=1.
- jpn_2015_7ab8a2 — 2015; 9; pipeline supplement; text; S=1.
- jpn_2015_bd227a — 2015; 7; pipeline supplement; text; S=0.
- jpn_2015_d092b4 — 2015; 4; pipeline supplement; text; S=0.
- jpn_2019_000cfe — 2019; 3; pipeline supplement; text; S=1.
- jpn_2019_13b27b — 2019; 6; pipeline supplement; text; S=1.
- jpn_2019_1de558 — 2019; 2; pipeline supplement; text; S=1.
- jpn_2019_a2033b — 2019; 9; pipeline supplement; text; S=1.
- jpn_2019_e2fec2 — 2019; 12; pipeline supplement; text; S=1.
- jpn_2021_0da367 — 2021; 5; pipeline supplement; pointer; S=0.
- jpn_2021_5230d7 — 2021; 2; pipeline supplement; text; S=1.
- jpn_2021_8a2d04 — 2021; 5; pipeline supplement; text; S=1.
- jpn_2021_aec1d9 — 2021; 9; pipeline supplement; text; S=1.
- jpn_2021_ce7bea — 2021; 4; pipeline supplement; text; S=1.
- jpn_2021_d82212 — 2021; 12; pipeline supplement; pointer; S=1.
- jpn_2022_135c78 — 2022; 8; pipeline supplement; text; S=1.
- jpn_2022_269940 — 2022; 1; pipeline supplement; text; S=1.
- jpn_2022_60aa31 — 2022; 10; pipeline supplement; text; S=1.
- jpn_2022_e8d416 — 2022; 9; pipeline supplement; text; S=1.

- jpn_2023_4980a0 — 2023; 3; pipeline supplement; text; S=1.
- jpn_2023_648f4e — 2023; 12; pipeline supplement; text; S=1.
- jpn_2023_87c2d1 — 2023; 2; pipeline supplement; text; S=1.
- jpn_2023_b42b43 — 2023; 9; pipeline supplement; text; S=1.
- jpn_2023_b7080f — 2023; 11; pipeline supplement; text; S=1.
- jpn_2023_cc8de7 — 2023; 1; pipeline supplement; pointer; S=0.
- jpn_2023_ccf086 — 2023; 8; pipeline supplement; pointer; S=0.
- jpn_2023_faa0e3 — 2023; 5; pipeline supplement; text; S=1.
- jpn_2024_5e3413 — 2024; 3; pipeline supplement; text; S=1.
- jpn_2024_851b05 — 2024; 4; pipeline supplement; text; S=1.
- jpn_2024_8d022e — 2024; 10; pipeline supplement; text; S=1.
- jpn_2024_e48072 — 2024; 1; pipeline supplement; text; S=1.
- jpn_2025_3422c1 — 2025; 3; pipeline supplement; text; S=1.
- jpn_2025_42febe — 2025; 7; pipeline supplement; text; S=1.
- jpn_2025_957b3e — 2025; 10; pipeline supplement; text; S=1.
- jpn_2025_efabb8 — 2025; 11; pipeline supplement; pointer; S=1.
- jpn_2025_fef72c — 2025; 1; pipeline supplement; pointer; S=0.

KMO

- kor_2006_38a9dc — 2006; 6; pipeline supplement; text; S=1.
- kor_2006_4bfe96 — 2006; 3; pipeline supplement; text; S=1.
- kor_2007_5b3a67 — 2007; 6; pipeline supplement; text; S=1.
- kor_2007_87b2ba — 2007; 2; pipeline supplement; pointer; S=1.
- kor_2008_d12348 — 2008; 6; pipeline supplement; text; S=1.
- kor_2009_03f95a — 2009; 5; pipeline supplement; text; S=1.
- kor_2009_89116d — 2009; 3; pipeline supplement; text; S=1.
- kor_2010_1dc4be — 2010; 5; pipeline supplement; text; S=1; checked W22.
- kor_2010_8787ab — 2010; 3; pipeline supplement; text; S=1.
- kor_2011_33308c — 2011; 6; pipeline supplement; text; S=1.
- kor_2011_ea207d — 2011; 3; pipeline supplement; text; S=1.
- kor_2012_7f29e1 — 2012; 6; pipeline supplement; text; S=1.
- kor_2012_e426b1 — 2012; 3; pipeline supplement; text; S=1.
- kor_2013_090453 — 2013; 5; pipeline supplement; text; S=1.
- kor_2013_3747e5 — 2013; 3; pipeline supplement; text; S=1.
- kor_2013_cb4e0d — 2013; 6; pipeline supplement; text; S=1.
- kor_2014_2f8031 — 2014; Problem 6; pipeline supplement; text; S=1.
- kor_2014_ef7b46 — 2014; Problem 3; pipeline supplement; text; S=1.
- kor_2016_12c7ef — 2016; Problem 6; pipeline supplement; text; S=1.
- kor_2016_3ddc3f — 2016; Problem 2; pipeline supplement; text; S=1.
- kor_2023_878c28 — 2023; 5; pipeline supplement; text; S=1.
- kor_2024_10ef8b — 2024; Problem 2; pipeline supplement; text; S=1.
- kor_2024_6e49fd — 2024; Problem 5; pipeline supplement; text; S=1.
- kor_2024_7a75e8 — 2024; Problem 3; pipeline supplement; text; S=1.
- kor_2024_89b8c1 — 2024; Problem 6; pipeline supplement; text; S=1.

OUT-OF-SCOPE

- mkd_2008_27a12f — 2008; 2; quarantine; pointer; S=0.

- mkd_2008_2a4984 — 2008; 4; quarantine; pointer; S=0.
- mkd_2010_d65b38 — 2010; Problem 3.; quarantine; pointer; S=0.
- mkd_2012_041e11 — 2012; 2; quarantine; pointer; S=0.
- esp_2013_1db4a4 — 2013; 1; quarantine; pointer; S=0.
- esp_2013_aee935 — 2013; 4; quarantine; pointer; S=0.
- esp_2014_4d971e — 2014; 1; quarantine; pointer; S=0.
- esp_2014_543539 — 2014; 5; quarantine; pointer; S=0.
- mkd_2014_5b64b3 — 2014; 1; quarantine; pointer; S=0.
- mkd_2015_4cea55 — 2015; 3; quarantine; pointer; S=0.
- mkd_2016_211258 — 2016; 2.; quarantine; pointer; S=0.
- mkd_2017_e39b7f — 2017; 5; quarantine; pointer; S=0.
- mkd_2018_93194d — 2018; 2; quarantine; pointer; S=0.
- sau_2022_065f88 — 2022; 3; quarantine; pointer; S=0.
- sau_2022_089603 — 2022; 83; quarantine; pointer; S=0.
- sau_2022_0cdea1 — 2022; 12; quarantine; pointer; S=0.
- sau_2022_116bdf — 2022; 23; quarantine; pointer; S=0.
- sau_2022_123706 — 2022; 4; quarantine; pointer; S=0.
- sau_2022_1f2124 — 2022; 46; quarantine; pointer; S=0.
- sau_2022_263bb6 — 2022; 14; quarantine; pointer; S=0.
- sau_2022_306561 — 2022; 49; quarantine; pointer; S=0.
- sau_2022_31e32d — 2022; 98; quarantine; pointer; S=0.
- sau_2022_3ad2d4 — 2022; 20; quarantine; pointer; S=0.
- sau_2022_3fd890 — 2022; 97; quarantine; pointer; S=0.
- sau_2022_508fcd — 2022; 59; quarantine; pointer; S=0.
- sau_2022_78c1e9 — 2022; 40; quarantine; pointer; S=0.
- sau_2022_7fac2d — 2022; 4; quarantine; pointer; S=0.
- sau_2022_8762d6 — 2022; 44; quarantine; pointer; S=0.
- sau_2022_8c7ca0 — 2022; 33; quarantine; pointer; S=0.
- sau_2022_9772c3 — 2022; 96; quarantine; pointer; S=0.
- sau_2022_9ee59a — 2022; 8; quarantine; pointer; S=0.
- sau_2022_a70771 — 2022; 100; quarantine; pointer; S=0.
- sau_2022_b72bfb — 2022; 7; quarantine; pointer; S=0.
- sau_2022_b818b9 — 2022; 55; quarantine; pointer; S=0.
- sau_2022_bd7b05 — 2022; 34; quarantine; pointer; S=0.
- sau_2022_c1d7c2 — 2022; 3; quarantine; pointer; S=0.
- sau_2022_c257df — 2022; 61; quarantine; pointer; S=0.
- sau_2022_c54abc — 2022; 54; quarantine; pointer; S=0.
- sau_2022_c5a758 — 2022; 91; quarantine; pointer; S=0.
- sau_2022_d922ec — 2022; 67; quarantine; pointer; S=0.
- sau_2022_e22d5f — 2022; 5; quarantine; pointer; S=0.
- sau_2022_e6619f — 2022; 43; quarantine; pointer; S=0.
- sau_2022_ee3336 — 2022; 7; quarantine; pointer; S=0.
- sau_2022_f9e7bb — 2022; 3; quarantine; pointer; S=0.
- mkd_2023_1d90f4 — 2023; 5; quarantine; pointer; S=0.
- mkd_2023_b3e2ef — 2023; 3; quarantine; pointer; S=0.
- sau_2023_0973d8 — 2023; Problem 8.; quarantine; pointer; S=0.
- sau_2023_0de8fd — 2023; Problem 4.; quarantine; pointer; S=0.
- sau_2023_1a95ab — 2023; Problem 2.; quarantine; pointer; S=0.
- sau_2023_23a263 — 2023; Problem 3.; quarantine; pointer; S=0.

- sau_2023_2b5cc3 — 2023; Problem 13.; quarantine; pointer; S=0.
- sau_2023_350e3e — 2023; Problem 2.; quarantine; pointer; S=0.
- sau_2023_407393 — 2023; Problem 3.; quarantine; pointer; S=0.
- sau_2023_5699ff — 2023; Problem 2.; quarantine; pointer; S=0.
- sau_2023_56ffb5 — 2023; Problem 4.; quarantine; pointer; S=0.
- sau_2023_714626 — 2023; Problem 3.; quarantine; pointer; S=0.
- sau_2023_7df256 — 2023; Problem 1.; quarantine; pointer; S=0.
- sau_2023_7f3b0d — 2023; Problem 39.; quarantine; pointer; S=0.
- sau_2023_81350a — 2023; Problem 15.; quarantine; pointer; S=0.
- sau_2023_87adf1 — 2023; Problem 1.; quarantine; pointer; S=0.
- sau_2023_933122 — 2023; Problem 42.; quarantine; pointer; S=0.
- sau_2023_abdc66 — 2023; Problem 4.; quarantine; pointer; S=0.
- sau_2023_b0bf31 — 2023; Problem 1.; quarantine; pointer; S=0.
- sau_2023_ca3253 — 2023; Problem 10.; quarantine; pointer; S=0.
- sau_2023_d38057 — 2023; Problem 4.; quarantine; pointer; S=0.
- sau_2023_d830b5 — 2023; Problem 1.; quarantine; pointer; S=0.
- sau_2023_e3605b — 2023; Problem 6.; quarantine; pointer; S=0.
- sau_2023_f3650b — 2023; Problem 3.; quarantine; pointer; S=0.
- sau_2023_f5bb58 — 2023; Problem 4.; quarantine; pointer; S=0.

RMM

- rou_2011_62bc85 — 2011; 6; contest; pointer; S=0.
- rou_2011_a9ff7f — 2011; 5; contest; pointer; S=0.
- rou_2011_bf1899 — 2011; 2; contest; pointer; S=0.
- rmm_2012_704e6f — 2012; Problem 5; contest; text; S=1; checked W14.
- rmm_2012_b79110 — 2012; Problem 1; contest; text; S=3; checked W02.
- rmm_2013_3e4b91 — 2013; Problem 6; contest; text; S=1.
- usa_2013_de41b7 — 2013; 5.; contest; text; S=2.
- rmm_2015_2cd14a — 2015; Problem 2.; contest; text; S=1.
- rmm_2015_4db6ce — 2015; Problem 6; contest; text; S=1.
- rmm_2015_d7c58c — 2015; Problem 3.; contest; text; S=1.
- rou_2015_4f438f — 2015; Problem 2; contest; text; S=1.
- rou_2015_eb480f — 2015; Problem 6; contest; text; S=1.
- rou_2016_185273 — 2016; Problem 6.; contest; text; S=1.
- rou_2016_dce07f — 2016; Problem 2.; contest; text; S=2.
- rou_2017_a94581 — 2017; Problem 5; contest; text; S=0.
- rou_2017_f8413b — 2017; Problem 3; contest; text; S=1; checked W25.
- rou_2018_53b77e — 2018; 5; contest; text; S=1.
- rou_2018_a59bad — 2018; 3; contest; text; S=1; checked W10.
- rmm_2019_0d3e45 — 2019; Problem 1; contest; text; S=1.
- rmm_2019_39ca85 — 2019; Problem 6; contest; text; S=0.
- rmm_2019_80fa3b — 2019; Problem 4; contest; text; S=0.
- rmm_2019_8e6703 — 2019; Problem 3; contest; text; S=1.
- rou_2020_2530c6 — 2020; Problem 3.; contest; pointer; S=0; checked W06.
- rou_2020_31b81f — 2020; Problem 4.; contest; pointer; S=0.
- rou_2020_5bb69d — 2020; Problem 5.; contest; pointer; S=0.
- rou_2020_66a457 — 2020; Problem 2.; contest; pointer; S=0.
- rmm_2021_03b2ae — 2021; Problem 4; contest; text; S=2; checked W26.

- rmm_2021_7ba909 — 2021; Problem 3; contest; text; S=1.
- rmm_2021_d11caf — 2021; Problem 2; contest; text; S=1.
- rou_2023_0cea08 — 2023; 6; contest; text; S=2; checked W17.
- rou_2023_9c5b23 — 2023; 2; contest; text; S=1.
- rou_2024_c3cc94 — 2024; Problem 3.; contest; text; S=1.
- rou_2024_d65b67 — 2024; Problem 1.; contest; text; S=3; checked W09.
- rou_2024_e309ed — 2024; Problem 6.; contest; text; S=1.
- rou_2024_eb35fe — 2024; Problem 2.; contest; text; S=1.
- rou_2025_742b0e — 2025; Problem 3.; contest; pointer; S=0.
- rou_2025_b5cc5a — 2025; Problem 6.; contest; pointer; S=0.
- rou_2025_e406cf — 2025; Problem 1.; contest; pointer; S=0.

UNRESOLVED

- mng_2007_76dd0e — 2007; ??-5; quarantine; pointer; S=0.
- mng_2012_0dd53f — 2012; II-3; quarantine; pointer; S=0.
- mng_2012_2873ae — 2012; ??????? 6; quarantine; pointer; S=0.
- mng_2012_359496 — 2012; I-2; quarantine; pointer; S=0.
- mng_2012_c3d4b9 — 2012; I-1; quarantine; pointer; S=0.
- mng_2012_d64f50 — 2012; ??????? 3; quarantine; pointer; S=0.

USAMO

- usa_2003_9ab1c6 — 2003; 6; contest; pointer; S=0.
- usa_2003_cc254a — 2003; 1; contest; pointer; S=0.
- usa_2003_fb5f08 — 2003; 3; contest; pointer; S=0.
- usa_2006_2edd72 — 2006; 3; contest; text; S=1.
- usa_2006_9454b3 — 2006; 2; contest; text; S=1; checked W19.
- usa_2006_ce5644 — 2006; 5; contest; text; S=3.
- usa_2007_83a6e7 — 2007; 1; contest; text; S=3.
- usa_2007_92862a — 2007; 3; contest; text; S=1.
- usa_2007_9951a2 — 2007; 3; contest; text; S=1.
- usa_2007_b62228 — 2007; 5; contest; text; S=1.
- usa_2007_f4d07c — 2007; 4; contest; pointer; S=1.
- usa_2008_2c2856 — 2008; 4; contest; text; S=0.
- usa_2008_6039a6 — 2008; 5; contest; text; S=1.
- usa_2008_8434eb — 2008; 6; contest; text; S=2.
- usa_2008_fddb53 — 2008; 3; contest; text; S=0.
- usa_2009_ddbe0d — 2009; 2; contest; text; S=1.
- usa_2010_30bc30 — 2010; 2; contest; text; S=2.
- usa_2010_83c87b — 2010; 3; contest; text; S=1.
- usa_2010_94fe39 — 2010; 6; contest; text; S=1.
- usa_2011_2951fb — 2011; 6; contest; text; S=1.
- usa_2011_75d863 — 2011; 2; contest; text; S=1.
- usa_2012_06c2de — 2012; 6.; contest; text; S=1.
- usa_2012_9b4f1a — 2012; 2; contest; text; S=1.
- usa_2013_070dad — 2013; 2.; contest; text; S=4.
- usa_2013_a35b16 — 2013; 3.; contest; text; S=1.
- usa_2014_952bd1 — 2014; 6.; contest; text; S=1.

- usa_2014_f6daba — 2014; 4.; contest; text; S=0.
- usa_2021_5d6120 — 2021; USAMO 4; contest; pointer; S=0.
- usa_2021_66def9 — 2021; USAMO 2; contest; pointer; S=0.
- usa_2021_8dcdf3 — 2021; USAMO 3; contest; pointer; S=0.
- usa_2025_a5e142 — 2025; USAMO 3; contest; text; S=0.
- usa_2025_e0957e — 2025; 5; contest; text; S=0.
- usa_2025_e518f7 — 2025; 6; contest; text; S=0.

VMO

- vnm_2001_4665d8 — 2001; 6; contest; text; S=1.
- vnm_2002_ddb25c — 2002; Problem 3; contest; text; S=1.
- vnm_2003_6ad236 — 2003; 10; contest; text; S=1.
- vnm_2003_98ada4 — 2003; 8; contest; text; S=1.
- vnm_2005_05160d — 2005; Problem 3.; contest; text; S=1.
- vnm_2006_4c4381 — 2006; Problem 3; contest; pointer; S=1.
- vnm_2006_fbb149 — 2006; Problem 6; contest; text; S=1.
- vnm_2007_eb910e — 2007; 4; contest; text; S=1.
- vnm_2008_0c937e — 2008; 5; contest; text; S=0.
- vnm_2008_6bc7d0 — 2008; 6; contest; text; S=0.
- vnm_2008_e8616d — 2008; 5; contest; text; S=0.
- vnm_2009_bd6666 — 2009; Problem 5; contest; text; S=0.
- vnm_2010_2bc522 — 2010; 5; contest; text; S=1.
- vnm_2010_8c54a3 — 2010; 4; contest; text; S=1.
- vnm_2011_accee2 — 2011; 4; contest; text; S=1.
- vnm_2012_3fa159 — 2012; Problem 4.; contest; text; S=0.
- vnm_2012_626786 — 2012; 1; contest; text; S=1.
- vnm_2012_736692 — 2012; Problem 6.; contest; text; S=1.
- vnm_2012_db0487 — 2012; 5; contest; text; S=1.
- vnm_2012_de8d74 — 2012; Problem 4.; contest; text; S=1.
- vnm_2012_e6adf8 — 2012; Problem 7.; contest; text; S=1.
- vnm_2012_ed74b6 — 2012; 4; contest; text; S=1.
- vnm_2013_1b2564 — 2013; Problem 1.; contest; text; S=1.
- vnm_2013_4017e2 — 2013; Problem 3.; contest; text; S=0.
- vnm_2013_6fb2bd — 2013; Problem 3.; contest; text; S=0.
- vnm_2013_8efd6e — 2013; 4; contest; text; S=1.
- vnm_2013_986218 — 2013; Problem 5.; contest; text; S=1.
- vnm_2013_a73731 — 2013; Problem 3.; contest; text; S=0.
- vnm_2013_c284da — 2013; Problem 6.; contest; text; S=0.
- vnm_2013_f344b9 — 2013; Problem 4.; contest; text; S=0.
- vnm_2014_a32c74 — 2014; 3; contest; text; S=1.
- vnm_2014_aef204 — 2014; 7; contest; text; S=1.
- vnm_2015_5ce2b3 — 2015; 7; contest; text; S=1.
- vnm_2015_7ffead — 2015; 7; contest; text; S=1.
- vnm_2015_ef4208 — 2015; 3; contest; text; S=1.
- vnm_2016_c11c43 — 2016; 4; contest; text; S=1; checked W21.
- vnm_2018_949f1a — 2018; 4; contest; text; S=1.
- vnm_2018_b1f4be — 2018; 3; contest; text; S=0.
- vnm_2018_da4108 — 2018; Problem 3; contest; text; S=0.

- `vnm_2018_dcf812` — 2018; 5; contest; text; S=1.
- `vnm_2019_e93538` — 2019; 7; contest; text; S=0.
- `vnm_2020_1719a9` — 2020; 7; contest; text; S=1.
- `vnm_2021_384922` — 2021; 6; contest; text; S=1.
- `vnm_2021_6a7ac6` — 2021; 1; contest; text; S=1.
- `vnm_2022_a59d06` — 2022; Problem 6; contest; text; S=1.
- `vnm_2023_423ad5` — 2023; 6; contest; text; S=1.
- `vnm_2024_9e7ac6` — 2024; 7; contest; text; S=1.
- `vnm_2024_b86eb8` — 2024; 5; contest; text; S=1.